Paediatric Urology 1

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 41	Management Strategies for Vesicoureteral Reflux
Ch 49	Anomalies of the Kidney and Collecting System
Ch 50	Renal Dysgenesis and Cystic Disease of the Kidney
Ch 51	Pathophysiology of Urinary Tract Obstruction
Ch 52	Ectopic Ureter, Ureterocele and Ureteral Anomalies
Ch 53	Surgery of the Ureter in Children
Ch 56	Hypospadias
Ch 57	Etiology, Diagnosis and Management of the Undescended Testis
Ch 61	Urologic Considerations in Pediatric Renal Transplantation
Ch 63	Paediatric Urologic Oncology: Renal and Adrenal
Ch 64	Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia
★	Putting it together: common paediatric decisions

41 Management Strategies for Vesicoureteral Reflux

1 · Definition and concept

We treat the infection risk, not the reflux.

- **Vesicoureteral reflux (VUR)** is retrograde urine flow to the upper tract, linked to **acquired scarring** and **congenital dysplasia**. It is a spectrum needing risk-stratified care.
- **Infection, not reflux, is the injury.** VUR carries infected urine to the kidney, so aim for sterile urine and good emptying.

2 · Embryology and antireflux mechanism

One fault explains reflux plus dysplasia.

- P1 P4 Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management — *Mackie–Stephens bud position (early bud gives high lateral orifice, VUR and dysplasia; late bud gives ectopia), passive flap valve, Paquin **5:1** vs 1.4:1*

Not in the paper answer:

- Abnormal bud contact also causes hypoplasia, so some "scars" are congenital.
- Opening is active (longitudinal tunnel muscle shortens and widens the tunnel); closure is passive.

3 · Primary vs secondary VUR

Secondary reflux needs the bladder treated first.

- P4 VUR: etiopathogenesis, recent trends in management and STING technique — *primary (deficient UVJ) vs secondary (PUV, neurogenic, dysfunctional voiding, ureterocele, duplication, diverticulum); overlap; detrusor overactivity commonest urodynamic finding; infant male high voiding pressures*

Not in the paper answer:

- Stricture and iatrogenic injury are other secondary causes.
- Infant male reflux comes from poor sphincter relaxation and settles as voiding matures.
- In spina bifida, VUR occurs in up to **30%**: start early clean intermittent catheterisation (CIC) and an anticholinergic or β3-agonist. Seek occult dysraphism: sacral dimple, hairy patch, gluteal cleft anomaly, refractory constipation.
- Normal bladders void at ~100 cm H_2O without reflux. **Reflux at low pressure or in early filling means poor resolution.**
- Treating detrusor overactivity (timed voiding, biofeedback, anticholinergics, α-blockers) improves resolution.

> **KEY POINT**
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Remove it and only true primary VUR remains.

4 · Epidemiology

Younger children with urinary tract infection (UTI) reflux more.

- VUR is found in **~30%** of children with UTI and ~17% of those imaged for hydronephrosis. It is present in **38.6%** under 2 years versus 7.6% at 12–21 years.

- Girls are affected ~2× overall, but infant boys predominate (up to 3:1 at 0–6 months).

5 · Grading — International Reflux Study (1981)

Grade drives every decision.

Grade	Ureter	Pelvis and calyces
I	Non-dilated, ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild–moderate dilatation	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation
V	Gross dilatation and tortuosity	Gross dilatation; **papillary impressions lost**

- Grades I–II are non-dilating and III–V dilating. Interobserver agreement is poor (RIVUR readers agreed ~60%).
- Radionuclide cystography (RNC) uses 3 grades: 1 = I; 2 = II–III; 3 = IV–V.
- **Ureteral diameter ratio (UDR)** is the largest ureteric diameter in the false pelvis divided by the L1–L3 vertebral distance. It predicts resolution and breakthrough UTI (BUTI) better than grade.

6 · Diagnosis

Who needs a cystogram, and which one.

When to image after UTI

- **American Academy of Pediatrics (AAP) 2011**, febrile UTI at 2–24 months: do **ultrasound (US) first**. Do a voiding cystourethrogram (VCUG) only if US is abnormal or after a **second febrile UTI**.
- **National Institute for Health and Care Excellence (NICE)**: US, with VCUG mainly for atypical or recurrent UTI in younger children and none routinely over 3 years.
- **Top-down**: dimercaptosuccinic acid (DMSA) scan first, VCUG only if abnormal. A normal DMSA leaves a **<1%** chance of high-grade VUR, but misses ~1/3 of (mostly low-grade) VUR.

Cystographic methods

- P1 VCUG — *gold standard; gravity ≤70 cm H_2O; early study once urine sterile; cyclic VCUG +12–20%; filling-phase reflux poor prognosis; postvoid pelvic retention suggests UPJO; RNC and ceVUS alternatives*

Not in the paper answer:

- Early VCUG (~8 days) finds as much VUR as a delayed one.
- The NICE approach misses much VUR, even grade IV–V.
- Direct RNC (^{99m}Tc pertechnetate) is more sensitive for grades II–V but **misses grade I**. Indirect RNC avoids a catheter but needs a toilet-trained child (sensitivity 70–90%). Contrast-enhanced voiding urosonography (ceVUS) shows intrarenal reflux; its grading is not standardised.
- Suspect coexisting **ureterovesical junction (UVJ) obstruction** with a dilated proximal ureter, distal aperistaltic segment, dilute contrast and delayed drainage. It is a **relative contraindication to endoscopic injection**.
- Cystoscopy is used only just before open surgery. Uroflowmetry and postvoid residual (PVR), with or without electromyography (EMG), help; invasive urodynamics only if refractory.

Upper tract

- **Renal and bladder ultrasound (RBUS)** is the serial mainstay but cannot exclude VUR.
- **DMSA** is the gold standard for cortical defects (sensitivity 98%, specificity 92%). Acute photopenia may resolve; a defect that persists is a scar.

7 · Congenital dysplasia vs acquired scar

Only the acquired scar is preventable.

- P4 P1 VUR: etiopathogenesis, recent trends in management and STING technique — *dysplasia (boy, IV–V, antenatal, not preventable) vs acquired scar (girl, febrile UTI, BBD, preventable); polar compound papillae, intrarenal reflux at 2 mmHg; Ransley–Risdon sterile reflux; "big bang" debated*

Not in the paper answer:

- Dysplasia gives a small, globally poor kidney; a scar gives focal photopenic defects.
- Sterile VUR damages only at high pressure, e.g. posterior urethral valves (Hodson).
- Scarring risk is highest in the first year.
- In prenatally detected boys, damage is confined to IV–V (nearly half of grade V), before any infection.
- Hypertension develops in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure (BP)**; removing a small kidney or segment may.

8 · Bladder and bowel dysfunction (BBD)

BBD is the main modifiable factor.

- **Treat constipation first**; it causes sphincter co-contraction.
- The American Urological Association (AUA) calls BBD a critical modifiable variable, bringing more UTI on prophylaxis and after surgery, less resolution, and **lower endoscopic success, but not open surgical success**.
- Of recurrent UTIs off prophylaxis, ~1/2 have BBD and VUR, ~1/5 VUR alone, ~1/3 BBD alone.
- **Phimosis**: in boys with VUR, circumcision cuts the number needed to treat (NNT) to prevent one UTI from ~111 to **~4**. Topical steroid is equivalent.

9 · Natural history and resolution

Most low-grade reflux resolves.

Grade	Spontaneous resolution
I–II	Most; grade II 63–85%
III	~50%; ~90% of resolutions within 4 years
IV–V, bilateral III	Few (infants with IV–V resolve well in year 1)

- Resolution favours **low grade, young age, unilateral reflux, no BBD, voiding-only reflux, higher bladder volume at onset and low UDR**. Age matters as much as grade (AUA 1997).

10 · Medical management — continuous antibiotic prophylaxis (CAP)

CAP failure is the main trigger for surgery.

- P3 Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome — *CAP agents by age with doses (TMP/amoxicillin <2 months; TMP-SMX, nitrofurantoin >2*

months), Hidas risk groups, indications for correction, 8–10% febrile UTI after stopping CAP

Not in the paper answer:

- Smellie's rule and the drug side-effect profiles (below).
- **Manage the infection risk, not the VUR.** Give a low dose **at night**, the longest retention.
- Trimethoprim-sulfamethoxazole (TMP-SMX): the sulfonamide displaces bilirubin (neonatal jaundice); rare reversible leukopenia. Nitrofurantoin causes the least faecal-flora resistance; pulmonary fibrosis and hepatotoxicity are rare at low dose.
- **Smellie's rule**: if the BUTI organism is sensitive to the prophylactic, suspect non-adherence or a low dose; if resistant, suspect a high residual or the dose.
- **If febrile UTI breaks through CAP despite good bladder and bowel habits, stop medical management and correct the VUR** (confirm the urine collection first).

Risk stratification (Hidas)

Factor	Low risk	High risk
Sex / age	Male; >5 y	Female; <1 y
Laterality / grade	Unilateral; I–II	Bilateral; III–V
BBD / scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

- Two-year BUTI on CAP is **8.6% low, 26% intermediate, 62.5% high** risk. ~2/3 of children are low risk.

11 · Practical algorithm (AUA-based, risk-stratified)

The clinic sequence.

ALGORITHM

1. Confirm true febrile UTIs; do RBUS. If multiple febrile UTIs, admissions or upper-tract findings, then VCUG. If afebrile or doubtful UTI with normal US, then conservative care.
2. In all, treat BBD (constipation first), with hygiene and timed voiding.
3. If low risk (grade I–II, no BBD, male, older), then observe with or without CAP, and treat any febrile UTI promptly.
4. If high risk (grade ≥III, girl, BBD, febrile presentation, scars, infant), then give CAP.
5. If breakthrough febrile UTI, new scars, high grade with poor function, or persistent VUR with low infection tolerance, then correct (endoscopic or reimplant).

- CAP helps little at grade ≤II without BBD, but helps at ≥III, at least in girls. Scarring, poor function or dysplasia favour correction.
- Unresolved: asymptomatic bilateral high-grade VUR in older boys; correction in girls before pregnancy.

12 · Surgical correction

Surgery stops reflux but does not protect kidneys better.

- Open correction succeeds in **>95%**; robotic correction is probably less morbid. Neither beats medical therapy for scarring or growth (IRSC, Birmingham).
- After correction almost no new scars form, but the adult UTI risk persists.

- **Endoscopic dextranomer/hyaluronic acid (Dx/HA)**: BBD lowers success. Avoid it with UVJ obstruction or obstructive refluxing megaureter. Mounds calcify in over half and **mimic ureteric stones**.
- Techniques: *Ch 53*. After a cross-trigonal reimplant, **retrograde catheterisation and ureteroscopy are difficult**.

13 · Associated conditions

Each anomaly changes the surgery.

- **Ureteropelvic junction obstruction (UPJO) with VUR**: dilated ureter, little or dilute pelvic contrast, no postvoid pelvic drainage. **Correct the obstruction first**; repairing both at once risks the ureteric blood supply.
- In a **duplex** kidney, VUR usually enters the **lower pole** (Weigert–Meyer); duplication alone does not raise risk.
- A **paraureteral diverticulum** resolves like primary VUR; a **ureter entering the diverticulum** lacks muscle backing and will not resolve.
- With a multicystic dysplastic kidney (MCDK), contralateral VUR (~1/4) usually resolves.
- **Megacystis–megaureter**: massive VUR recycles urine, so the bladder is large and thin; the urethra is normal. Treat by reimplant, not outlet surgery.
- **Obstructive refluxing megaureter**: ureteric contrast is more dilute than bladder. Avoid injection; reimplant with or without tapering.

14 · Sibling screening

Screening seeks damaged kidneys, not every reflux.

- Sibling prevalence is ~**30%** (up to 50%), higher when younger.
- The target is **renal damage, not VUR**. AUA: **VCUG for siblings with a cortical abnormality or renal asymmetry on US, or a UTI history**; otherwise observe.

15 · Follow-up and adulthood

Scars carry lifelong risk.

- If scarred, check BP, proteinuria and renal function for life.
- **Pregnancy**: VUR leads to pyelonephritis in ~1/3. Scars bring ~3× hypertension and **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.** Reimplanted women keep a UTI risk but no excess miscarriage.

16 · Landmark trials

Why prophylaxis is now selective.

Trial	Design	Result	Take-home
IRSC	<9 y, high grade; CAP vs open surgery	UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal	Surgery ≈ medical for scarring
Birmingham Reflux Study	High grade; medical vs surgical, 5 y	New scars equal, all within 2 y	Equal; supports "big bang"

Trial	Design	Result	Take-home
Swedish Reflux Trial	1–2 y, grade III–IV; CAP vs endoscopic vs surveillance	Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; no new scars on CAP; boys no difference	Treat girls with dilating VUR
RIVUR	607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo	Recurrent UTI **12.9% vs 23.6%**; scarring unchanged; most benefit with febrile index UTI/BBD	CAP prevents UTI, not scars
PRIVENT	576 children with UTI; CAP vs placebo	Absolute UTI reduction 6% (from 19% to 13%)	Modest benefit
French / low-risk trials	Grade I–III; CAP vs none	No significant difference	No benefit at low risk

KEY POINT

Trials disagree because of population risk: RIVUR was diluted by low-risk children (~8% grade IV); reanalysis gives NNT 5 high-risk vs 18 low-risk. CAP cuts infection in proportion to baseline risk; neither CAP nor surgery prevents scarring better.

49 Anomalies of the Kidney and Collecting System

1 · Embryology that explains everything

Every anomaly here is an error of this one process.

- The ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; its tips induce nephrons (**reciprocal induction**).
- In weeks 6–9 the kidney ascends and rotates 90° ventromedially: calyces lateral, pelvis medial.
- Renal and genital development are linked, so renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies. The adrenal (cortex mesoderm, medulla neural crest) is independent.

KEY POINT
Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

2 · Bilateral renal agenesis (BRA)

BRA is uniformly lethal, so recognise it and counsel the family.

- It occurs in **≈1 in 5000** births, **75% male**. First-degree relatives carry ≈1000× risk, so **scan parents and siblings by ultrasound (US)** of any infant with unilateral agenesis (URA), BRA or dysgenesis.
- **Pathology**: ureters are atretic; the bladder is absent (≈½) or hypoplastic, because UB/WD entry into the cloaca starts bladder development. The flat adrenal gives the **"lying down" sign**.
- **Potter facies** (senile look, infra-canthal fold, blunted nose, low-set-appearing ears, claw hands, bowed clubbed legs) are **deformations** from absent amniotic fluid, not malformations.
- Associations: undescended testes 43%, testicular agenesis ≈10% (vas usually normal), female müllerian/ovarian anomalies, gastrointestinal anomalies ≈60%.
- **Pulmonary hypoplasia** is the cause of death. Fetal urine forms >90% of third-trimester amniotic fluid. **Peters two-step model**: a kidney-derived "renal factor" (proline) acts before 16 wk, then amniotic fluid in the canalicular phase (16–26 wk), so fluid alone only partly restores the lungs.
- **Diagnosis**: prenatally, severe oligohydramnios, no kidneys and a small chest. In a neonate, **anuria >24 h without bladder distension** means BRA until proved otherwise. Use US with power Doppler; if equivocal, dimercaptosuccinic acid (DMSA) shows no uptake.
- **Prognosis**: ≈40% are stillborn and the rest die in 24–48 h of respiratory failure. Amnioinfusion is experimental (RAFT, §9): only under ethics-board (IRB) oversight, with full counselling including palliative care.

KEY POINT
BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

3 · Unilateral renal agenesis (URA) and the solitary functioning kidney (SFK)

Often silent; the issues are genital anomalies and lifelong hyperfiltration.

- URA occurs in **1 in 1200**, male 1.8:1, slightly more on the left. Many apparent URAs are an **involuted multicystic dysplastic kidney (MCDK)** (curvilinear calcification on computed tomography, CT; colonic flexure in normal place). Renal aplasia may be the commonest cause of congenital solitary kidney. At 20 weeks, adrenal or spleen may be mistaken for kidney.
- Associations include VACTERL (≈30% have URA), MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia), Turner, Kallmann, branchio-oto-renal and DiGeorge syndromes. **Maternal diabetes triples the risk**. An isolated preauricular pit or single umbilical artery needs no US; two or more anomalies do.
- **The ipsilateral ureter is absent in ≈60%**. "Hemitrigone" is a misnomer (the trigone comes from the urogenital sinus). The defect is in the UB (RET), so the gonad is rarely affected.
- **Contralateral kidney**: vesicoureteral reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction (VUJ) obstruction ≈7%.
- **Genital anomalies occur in 20–40%** (females 25–50%, males 10–15%). Adrenal agenesis occurs in <10%.

	Male (WD derivatives)	Female (MD derivatives)
Always present	Testis, head of epididymis	Fimbriated end of tube
Commonest	Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney**	Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter; infertility up to ⅓ with unicornuate uterus
Named syndrome	**Zinner**: seminal vesicle cyst and ipsilateral agenesis	**OHVIRA** (obstructed hemivagina and ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pain
Trigger to image kidneys	Impalpable vas (infertility, herniotomy/orchidopexy)	Any uterine anomaly; pelvic US after thelarche, before menarche

- **Magee classification**: type I (before wk 4, whole ridge) gives a unicornuate uterus; type II (early wk 4, WD and UB) gives didelphys with obstructed hemivagina on the agenesis side; type III (after wk 4, UB only) gives isolated URA, normal genitalia.
- **Mayer–Rokitansky–Küster–Hauser (MRKH)**: 46,XX with upper vaginal to total müllerian agenesis (1 in 4500 females). Type I rarely has renal anomalies. **Type II (commoner) has renal anomalies in 40–60%**. MURCS is the most severe end.
- **Work-up**: compensatory hypertrophy is seen from 20 wk in ≈90%. Use US and Doppler; DMSA finds a missed ectopic kidney. A voiding cystourethrogram (VCUG) for the 28% contralateral VUR is **not routine** (no fewer febrile urinary tract infections, UTIs); decide with the family.
- **Sport** is not precluded: assess individually and use padding (vehicle accidents cause more renal trauma).

Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass leads to glomerular hypertension, then albuminuria and sclerosis. Risk is highest when the SFK itself has a congenital anomaly of the

kidney and urinary tract (CAKUT). Markers: low birth weight, prematurity, small kidney, diabetic mother.
- In KIMONO, ≈⅓ had renal injury by ≈10 y, and population ESRD risk is ≈4×, so follow-up is lifelong.

Situation	Blood pressure (BP) and first-morning urine albumin	Ultrasound	Creatinine
SFK without CAKUT	Annually	Every 5 y if glomerular filtration rate (GFR) >60 and no medication	Periodic, into adolescence
SFK with CAKUT	Twice yearly	As indicated	Periodic
GFR <60 or on antihypertensive/antiproteinuric	BP, albumin, creatinine 2–4×/year		

Hypertension or microalbuminuria (>30 mg/24 h) calls for an angiotensin-converting enzyme (ACE) inhibitor, salt limitation and no excess protein. Hypoplasia and dysplasia: *Ch 50*.

4 · Supernumerary kidney

Rare; the point is to tell it from a duplex kidney.

- It has its **own capsule, collecting system and separate blood supply**, whereas a duplex kidney has one capsule and a shared supply. It is usually left-sided, ≈60% lie caudal to the dominant kidney, and the ureters join or stay separate (≈50% each).
- About ½ are hydronephrotic, and obstruction, infection or stones appear in early adulthood. Magnetic resonance urography (MRU) or retrograde study shows anatomy, renography function; cystoscopy shows one or two ipsilateral orifices.

5 · Ectopia (anomalies of ascent)

An ectopic kidney is easily mistaken for agenesis or a pelvic tumour.

Simple ectopia

- It occurs in **1 in 900** at autopsy and is bilateral in ≈10%. **Pelvic is commonest**, then lumbar, abdominal, cephalad and thoracic. Aberrant vessels are a consequence of ectopia, not its cause.
- The kidney is small, lobulated and malrotated (pelvis anterior). **Hydronephrosis is common**: ½ obstruction (PUJ ≈70%), ¼ VUR ≥ III, ¼ malrotation. **VUR occurs in 30–50%**, often into the orthotopic kidney.
- The blood supply is anomalous: **map it by CT or MR angiography (MRA) before surgery**.
- Genital anomalies occur in 15–45%: females 20–66%; in males, hypospadias and undescended testis ≈5% each. Ectopia occurs in ≈14% of cloacal malformations. Continuous incontinence in a girl suggests an ectopic ureter from the ectopic kidney.
- With an empty renal fossa, DMSA or MRU may find a small non-functioning kidney mistaken for URA. The other kidney is usually normal; the adrenal is normally placed.
- **Never remove a "pelvic mass" without imaging**: solitary pelvic kidneys have been excised as tumours.

Cephalad and thoracic kidney

- Cephalad ectopia is classically seen with **omphalocele**.

- A thoracic kidney (≈1 in 13,000) lies in the **foramen of Bochdalek**, outside the pleura, usually silent. Cause: delayed mesonephric involution; 0.25% of diaphragmatic hernias contain a kidney. A smooth posterior mass on chest X-ray needs CT or MRU, not thoracotomy; operate only for respiratory distress. The adjacent lower lobe may be hypoplastic.

6 · Fusion anomalies

Fused kidneys bring obstruction, stones and variable vessels.

Crossed renal ectopia

- The kidney lies opposite its ureteric orifice; **90% are fused**, making it the second commonest fusion anomaly. Left-to-right crossing is 3:1. The ureter crosses at the pelvic brim and its orifice is orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed (imperforate anus ≈20%) and bilateral crossed (rarest).

Crossed fused form	Key feature
Inferior ectopia (≈⅔, commonest)	Crossed upper pole fused to normal lower pole; pelves anterior
Sigmoid (S-shaped)	Second commonest; pelves face opposite ways
Lump (cake)	Lobulated pelvic mass
L-shaped (tandem)	Crossed kidney transverse, anterior to L4
Disc (pancake)	Medial fusion, reniform outline kept
Superior ectopia	Least common; crossed lower pole to normal upper pole

- An **anterior pelvis means early fusion** (before rotation); a medial pelvis means late fusion. The inferior mesenteric artery (IMA) and mesenteric root stop ascent.
- VUR occurs in ≈20% (71% in bilateral crossed) and an ectopic orifice in ≈3%. Genital anomalies include cryptorchidism, hypospadias, absent vas and vaginal atresia.
- Pain, UTI, haematuria and stones appear in the 3rd–4th decade. **A stone in the crossed ureter hurts on the anephric side**; pyelonephritis or PUJ obstruction hurts where the kidneys lie.
- Image with US first, diuretic renography for drainage, MRU or MRA before major surgery. A midline-fused kidney whose single ureter divides into two pelves goes with imperforate anus and vertebral anomalies.

Horseshoe kidney

- It is the **commonest fusion anomaly**: **1 in 400** (0.25%), male 2:1. **Lower-pole fusion occurs in 95%**, before rotation.
- The **isthmus** (parenchyma or a fibrous band) lies at L3/L4, **just below the IMA**, in front of the aorta and inferior vena cava (IVC).
- The calyces **point posteriorly**, with the lowest medial (it may overlie the spine). The ureter inserts high and bends over the isthmus. **The blood supply is highly variable**, so obtain MRA or CT before surgery.
- **Associations** occur in ≈½: **≈60% of Turner patients have a horseshoe kidney**; also caudal regression, neural tube defects (3%), anorectal/VATER anomalies, and hypospadias or undescended testis ≈4%.

- **PUJ obstruction occurs in 13–34%** (high insertion, ureter over isthmus, crossing vessels), but many hydronephroses are not obstructed. VUR occurs in 8–32%.
- **Stones affect ≈⅓ of adults** from stasis and metabolic risk, so still do a metabolic work-up.
- **Tumours**: **Wilms risk is 1.76–7.9×** (favourable histology); preoperative chemotherapy preserved ≈75% of parenchyma (National Wilms Tumor Study, NWTS). Pelvic tumours are increased; renal cell carcinoma is not.
- At least 50% are asymptomatic; pain is vague, abdominal to lower lumbar; UTI occurs in ≈30%. US may miss a fibrous isthmus, while DMSA shows a functioning band. Pregnancy is unaffected and the kidney is transplantable.

KEY POINT
† PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

7 · Malrotation and vascular anomalies

Vessel anatomy tells the rotation and guides surgery.

- Malrotation occurs in ≈1 in 939, male 2:1, and often in Turner syndrome. **Weyrauch** types: **ventral (no rotation, commonest)**, ventromedial, dorsal (180°, rarest) and lateral (reverse). **The vessel course is the only clue** to direction.
- Hilar fibrosis causes intermittent PUJ obstruction and stones, but function is unaffected. Imaging: flattened elongated pelvis, stretched upper calyx, laterally displaced upper ureter. Suspect it when a stone lies in an unusual position. Bilateral malrotation mimics a horseshoe kidney.
- P2 Anomalies of renal artery. — *multiple / aberrant / accessory terms; single artery 71–85%; end-arteries; crossing vessels in normal PUJs; aneurysm (Abeshouse, treat >2 cm, pregnancy, symptoms); congenital vs acquired AVF, embolisation*

Not in the paper answer:

- Aneurysm: ring calcification ≈30%; hypertension is renin-mediated; rupture occurs only >3 cm.
- Congenital arteriovenous fistula (AVF) presents in the 3rd–4th decade: bruit ≈75%, haematuria >75%, hypertension and high-output failure ≈50%. It rarely resolves; balloon occlusion or nephrectomy are options beyond embolisation.
- Treat an aneurysm for rupture, growth, pregnancy, size **>2 cm** or symptoms. An acquired AVF may close within months, so observe it.

8 · Collecting system anomalies

The question for each one is whether there is obstruction.

- P1 Pelvicalyceal anomalies other than PUJ obstruction — *calyceal diverticulum (types I/II, stones ≈40%, delayed contrast pooling, percutaneous ablation and fulguration), hydrocalycosis, megacalycosis (non-obstructive, 12–20 calyces, male 6:1, no surgery), infundibulopelvic stenosis (bilateral, >⅓ renal failure), extrarenal and bifid pelvis (≈10%)*

Not in the paper answer:

- For a diverticulum, percutaneous ablation with fulguration lasts best; ablating the cavity lowers recurrence.
- Hydrocalycosis presents with pain, UTI or haematuria.
- Megacalycosis with a segmental distal megaureter can mimic obstruction.
- Bilateral megacalycosis occurs in males; only a mild concentrating defect results.

- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: check creatinine, estimated GFR (eGFR) and urinalysis yearly. If function falls, give an ACE inhibitor or calcium-channel blocker and restrict protein; plan surgery on 3-D MRU.

KEY POINT

Collecting-system anomalies split on one question — is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

9 · Landmark trials

Fetal therapy for BRA.

Trial	Design	Result	Take-home
RAFT (Renal Anhydramnios Fetal Therapy)	Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk	94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early	Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only

50 Renal Dysgenesis and Cystic Disease of the Kidney

1 · Genetics in one table

Four genes explain most syndromic dysgenesis and cystic disease.

Gene	Renal phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease

2 · Dysgenesis: dysplasia vs hypoplasia

Dysplasia is the leading cause of childhood renal failure.

- **Dysgenesis** is maldevelopment of size, shape or structure, and includes dysplasia and hypoplasia. Agenesis: *Ch 49*.
- **Dysplasia** is a **histological** diagnosis: metanephric differentiation arrests because of an error between the wolffian duct, ureteric bud (UB) and blastema. Its sine qua non is **primitive ducts** with fibromuscular collars; cartilage and immature glomeruli also occur. Not all cysts are dysplastic, and not all dysplasia is cystic.
- Causes are primary UB failure (Fraser, BOR, renal–coloboma, Kallmann, Townes–Brocks, HDR syndromes) or fetal outflow obstruction. **It is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US) shows a small echogenic kidney (brighter than liver), lost corticomedullary differentiation and sometimes cysts.
- **Hypoplasia** is reduced mass from too few cells; simple hypoplasia has a normal nephron number.

Entity	Defining feature	Course / management
Oligomeganephronia	Few, hypertrophied nephrons; usually bilateral	After year 1: polyuria, polydipsia, short stature; **proteinuria first sign**; rapid fall in teens, reaching ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, angiotensin-converting enzyme (ACE) inhibitor; living related donor acceptable (not familial)
Ask-Upmark (segmental hypoplasia)	Mostly acquired reflux nephropathy; deep "slit-scars", thyroid-like tubules	Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures blood pressure (BP) regardless of renin (persisting BP suggests a contralateral scar); bilateral: medical, dialysis/transplant. Reflux correction does not cure hypertension
Hypodysplasia	With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly	**Dysplasia is proportional to the degree of ureteric ectopia**: lateral orifice gives reflux-associated, medial/ureterocele gives obstruction-associated dysplasia. PUV: mild (subcapsular cysts) vs severe (cysts, cartilage). Prune-belly: tortuous ureters, usually no outlet obstruction

KEY POINT
The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

3 · Cystic disease: concepts and classification

Sort a cystic kidney first by inheritance, then by the extrarenal clue.

- Cysts arise from tubules through **epithelial proliferation** (a "second hit"), **fluid secretion** (net NaCl secretion) and **abnormal extracellular matrix**. The **primary cilium** is central.
- A **multicystic** kidney (MCDK) is dysplastic with no normal nephrons. A **polycystic** kidney starts normal, and its nephrons are progressively replaced by cysts.

Inheritable	Non-heritable
Autosomal recessive polycystic kidney disease (ARPKD)	Multicystic dysplastic kidney (MCDK)
Autosomal dominant PKD (ADPKD)	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant)	Simple cysts
Congenital nephrosis; familial hypoplastic glomerulocystic disease	Medullary sponge kidney; sporadic glomerulocystic disease
Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL)	Acquired renal cystic disease; calyceal diverticulum

The extrarenal clue decides: hepatic fibrosis means ARPKD; berry aneurysm, ADPKD; retinitis pigmentosa, nephronophthisis; angiofibroma, tuberous sclerosis complex (TSC); haemangioblastoma, VHL. Chromosome 16 carries PKD1 and TSC2.

4 · ARPKD

A neonatal disease of huge echogenic kidneys and an always-fibrotic liver.

- P3 Autosomal recessive polycystic kidney disease. — *PKHD1 chr 6, fibrocystin; always congenital hepatic fibrosis; oligohydramnios, pulmonary hypoplasia, 30–50% neonatal death; homogeneously hyperechoic kidneys; scan parents; supportive care; ARegPKD early bilateral nephrectomy warning*

Not in the paper answer:

- About ½ of neonatal survivors are alive at 10 y. In later-onset disease, hepatic problems dominate (portal hypertension with normal liver enzymes).
- On US the pyramids merge with the cortex. Use computed tomography (CT) if a tumour is suspected (Wilms, mesoblastic nephroma).
- Nephrectomy costs nephron mass (earlier dialysis) and risks hypotension after bilateral removal (also predicted neurological injury). The remaining kidney may grow fast after unilateral nephrectomy, forcing a second operation, so buy time.

5 · ADPKD — paediatric focus

In children the key finding is hypertension with normal function.

- P2 ADPKD (Adult polycystic kidney disease) — *PKD1 (16, 85%, ESRD 50s) vs PKD2 (4, 15%, ESRD 70s); pain, haematuria ≈50%, stones 20–30%, hypertension ≈50% at 20–35 y; liver cysts, aneurysm 10–30%; Ravine criteria; BP <130/80 with ACEi/ARB, tolvaptan (TEMPO, REPRISE), lipophilic antibiotics, unroofing*

Not in the paper answer:

- About 10% arise de novo, and in utero or infantile onset is aggressive. In PKD1, most children have cysts by 10 y; ≈17% look unilateral or asymmetric.
- **Contiguous gene syndrome**: deletion of PKD1 and the adjacent TSC2 gives TSC with severe early polycystic kidneys.
- Early ESRD risk: PKD1, male sex, haematuria <30 y, hypertension <35 y, hyperlipidaemia, sickle trait.
- A negative US in a young person does **not** exclude PKD2 (sensitivity ≈⅔ <30 y), so use genetic testing when certainty matters, after counselling.
- **With no family history**, diagnose on bilateral cysts plus ≥2 of: bilateral enlargement, ≥3 hepatic cysts, cerebral aneurysm, arachnoid/pineal/pancreas/spleen cyst.
- Screen for aneurysm by magnetic resonance angiography (MRA) only for family history, prior rupture, high-risk job, major surgery or anxiety. Observe <7 mm; coil or clip if >7–10 mm or symptomatic.
- **Hypertension with normal function is the key paediatric finding**. Tight control slows decline and lowers cardiac and aneurysm morbidity.
- Liver cysts are rare in children and worse in females. Renal cell carcinoma (RCC) is not increased.
- On CT a haemorrhagic cyst measures 50–90 Hounsfield units (HU); on magnetic resonance imaging (MRI) it is bright on T1 and dark on T2. Tolvaptan (approved 2018) causes aquaresis and hepatotoxicity. Stones are uric acid and oxalate equally; cysts camouflage hydronephrosis.

> **KEY POINT**
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

6 · Juvenile nephronophthisis (NPH) / medullary cystic kidney disease (MCKD)

A salt-wasting cause of childhood renal failure that ultrasound often misses.

- These ciliopathies show **tubular basement membrane thickening, tubular atrophy with corticomedullary cysts and interstitial fibrosis** in small-to-normal kidneys. **Polyuria and polydipsia occur in >80%** (vasopressin-resistant, with **salt wasting**). ESRD follows 5–10 y after presentation.

	Juvenile NPH	MCKD
Inheritance	Autosomal recessive	Autosomal dominant
ESRD	Mean 13 y; **10–20% of childhood renal failure**	Third–fourth decade
Hypertension	Absent	Present
Extrarenal	20%: **retinitis pigmentosa (Senior–Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl	None; type 2 — hyperuricaemia, gout in teenage males

- US often misses the cysts; CT or MRI is better. Bardet–Biedl syndrome is obesity, intellectual disability, polydactyly, retinitis pigmentosa and hypogenitalism.
- **Replace sodium, and avoid salt restriction and diuretics**. Transplant works, with no recurrence in grafts.

7 · Syndromes with renal cysts

In these syndromes the kidney lesions threaten life and function.

Tuberous sclerosis complex (TSC)

- It is autosomal dominant: **TSC1 (chr 9), TSC2 (chr 16)**. Epilepsy occurs in ≈80%, intellectual disability ≈60% and adenoma sebaceum ≈75%; the earliest skin sign is the ash-leaf macule. Diagnosis needs 2 major features, or 1 major and 2 minor (angiomyolipoma is major; multiple renal cysts minor).
- Kidney: cysts (mimic ADPKD) and **angiomyolipoma (AML) in 40–80%**, bilateral, multiple, after 10 y. AMLs never metastasise; the risk is haemorrhage. RCC occurs in ≈2% (younger, female, bilateral); suspect it in a growing fat-free or calcified lesion.
- At diagnosis, obtain abdominal MRI, BP and glomerular filtration rate (GFR). AML fat measures ≈−50 HU on CT. **MRI is preferred**, because US misses 25–30% of fat-poor AMLs; recognising fat-poor AML avoids surgery.
- **If an AML is >3 cm and growing, give everolimus (a mammalian target of rapamycin, mTOR, inhibitor) first line**. For an acute bleed, embolise and then give corticosteroids. Kidney-sparing surgery is second line; avoid nephrectomy. Image every 1–3 y (yearly if near 3 cm or growing); check BP, proteinuria and GFR yearly; treat hypertension with ACE inhibitor or angiotensin receptor blocker (ARB).

Von Hippel–Lindau disease

- P2 VHL gene and von Hippel–Lindau disease — *AD, 3p25, pVHL degrades HIF, which drives VEGF; lesion frequencies (cysts 76%, clear cell RCC, phaeo, haemangioblastoma, epididymal cystadenoma); diagnostic criteria; VHL Alliance 2020 surveillance (abdominal MRI 2-yearly from 15); nephron-sparing surgery, recurrence 75–85%, DSS 81–94%; partial adrenalectomy*

Not in the paper answer:

- Surveillance also includes examination yearly from 1 y and BP from 2 y. Stop routine imaging at 65, and do a full check before conception.
- Bilateral radical nephrectomy is only for high-grade disease; ablation is investigational.
- Partial adrenalectomy for a small phaeochromocytoma avoids lifelong steroids.

8 · Multicystic dysplastic kidney (MCDK)

The MCDK is harmless; watch the opposite kidney.

- It has non-communicating cysts, **no normal parenchyma, no function** and an atretic ureter ("bunch of grapes"); types are infundibulopelvic (commoner) and hydronephrotic.
- It is the **commonest renal cystic disease of the newborn and the second commonest neonatal abdominal mass**: 1 in 1000–4000, male 55–70%. Bilateral disease is lethal. The obstruction theory competes with the favoured **UB-gene theory** (EYA1, PAX2, WT1, GDNF), which explains contralateral anomalies. It rarely occurs in a duplex upper pole or a horseshoe moiety.
- **The contralateral kidney** hypertrophies and has **vesicoureteral reflux (VUR) in 18–43% and pelviureteric junction (PUJ) obstruction in 3–12%**.
- About 40% involute (later imaged as "agenesis"). Hypertension and Wilms tumour are not increased.

US feature	MCDK	Severe PUJ obstruction
Cysts	Haphazard, varied, non-communicating	Calyces around central pelvis, communicating

US feature	MCDK	Severe PUJ obstruction
Central large cyst	Absent	Present (pelvis)
Isotope uptake	None	Usually some

ALGORITHM

1. For a prenatally detected MCDK, do US on day 2–3 and at 1 month to assess the contralateral kidney and bladder.
2. If the diagnosis is in doubt, do a dimercaptosuccinic acid (DMSA) or mercaptoacetyltriglycine (MAG3) scan.
3. If there is a urinary tract infection (UTI) or an abnormal contralateral kidney or bladder, then do a voiding cystourethrogram (VCUG); otherwise do not.
4. Hand over to primary care to check BP, mass and UTI.

- **Nephrectomy** is only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance (3–12-monthly) has no proven benefit. The risk of malignant transformation is extremely low, perhaps nil.

KEY POINT

MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction. Mistaking a non-functioning PUJ kidney for MCDK costs little.

9 · Benign multilocular cyst (cystic nephroma)

Imaging cannot exclude cystic Wilms tumour, so these lesions are excised.

- Spectrum (no conversion): multilocular cyst, cystic partially differentiated Wilms (CPDN), cyst with Wilms nodules, cystic Wilms. The adult counterpart is multilocular cystic RCC.
- **It is bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into the pelvis)**.
- Imaging separates it from MCDK but not from cystic Wilms, mesoblastic nephroma or clear cell sarcoma, so **excision is both diagnosis and treatment** (nephrectomy in children; partial if localised). Wilms nodules are staged by National Wilms Tumor Study (NWTS) criteria.

10 · Simple cysts in children and Bosniak

When to ignore, follow or remove a cyst.

- **US criteria (all four)**: anechoic, thin smooth wall, posterior enhancement, round or oval. On CT a simple cyst is −10 to +20 HU without enhancement. A hyperdense (20–90 HU) cyst is benign if non-enhancing, homogeneous, ≤3 cm and ≥¼ of its wall is extrarenal. Calcification occurs in 2% of cysts (peripheral) vs 10% of RCC (central).
- Puncture is now rare: suspected infection, discordant imaging, borderline lesion in a poor surgical candidate.
- **Children**: if a simple cyst is <15 mm and asymptomatic, repeat US at 2 y and discharge if stable. Continue follow-up if it is ≥15 mm or complex, or with a family history or genitourinary anomalies. Do not label bilateral cysts without liver cysts as ADPKD without family history or genetics.

Bosniak	Features	Management
I	Simple	None
II	Fine septa, minimal calcification, hyperdense ≤3 cm	None
IIF	More septa/calcification, no measurable enhancement	Follow-up (≈11% progress)
III	Thick/irregular enhancing walls or septa	Excision (paediatric III–IV: >90% intermediate/malignant)
IV	Enhancing solid components	Excision

Treat a symptomatic benign cyst by aspiration with or without ethanol sclerosis, or by laparoscopic unroofing; drain an infected cyst and give antibiotics. Bosniak 2019 adds MRI and down-classifies lesions to curb over-treatment.

11 · Medullary sponge kidney, glomerulocystic and acquired cystic disease

These cause stones, bleeding or cancer rather than childhood renal failure.

- P2 Medullary sponge kidney — *ectatic collecting ducts, urographic triad ("brush bristles", "bouquet of flowers"), presents >20 y, colic 50–60%; hypercalciuria ⅓–½, incomplete dRTA 30–40%; potassium citrate first, thiazide, phosphate (not with urease infection)*
- **Sporadic glomerulocystic disease** gives bilateral enlarged kidneys with cysts in Bowman space; it is a diagnosis of exclusion.
- P2 Acquired renal cystic disease — *≥3 cysts/kidney in ESRD; uraemia not dialysis; 10% pre-dialysis, rising to >90% at 10 y; small kidneys; haemorrhage, adenomas, RCC (aggressive in native kidneys after transplant); bleeding: embolisation/nephrectomy; tumour >3 cm excise, <3 cm nephrectomy or yearly CT*

Not in the paper answer:

- Acquired renal cystic disease (ARCD) affects 20–45% of children on dialysis.
- A retroperitoneal bleed calls for nephrectomy, because an occult RCC is likely.
- Avoid iodinated contrast before dialysis.
- For heparin-related haematuria, switch to peritoneal dialysis.
- **Parapelvic (renal sinus) cysts** are lymphatic and mimic hydronephrosis on US. Delayed contrast CT shows an undilated, displaced system; manage conservatively.

12 · Landmark trials

Evidence for ADPKD treatment and ARPKD nephrectomy timing.

Trial	Design	Result	Take-home
HALT-PKD	Randomised: lisinopril and telmisartan vs lisinopril; rigorous vs standard BP	Dual blockade no benefit; low BP slowed total kidney volume (TKV) growth, lowered left ventricular mass and albuminuria, GFR unchanged	Single renin–angiotensin blocker; tight BP early
TEMPO / REPRISE	Randomised, tolvaptan vs placebo	Slower TKV growth and GFR decline	First disease-modifying drug; aquaresis, hepatotoxicity
ARegPKD registry	Observational ARPKD cohort	Bilateral nephrectomy <3 months gave the worst neurological outcome	Avoid early bilateral nephrectomy

51 Pathophysiology of Urinary Tract Obstruction

1 · Definition and classification

Function, not dilation, defines obstruction.

- **Obstruction** is any anatomic or functional impediment to flow, calyces to meatus, that injures the kidney or lower tract: **acute or chronic, partial or complete, intrinsic or extrinsic, unilateral or bilateral**.
- It is defined by **flow and function, not dilation**: partial versus high-grade, not the image, decides treatment.
- It affects ≈ **1 in 500** live births; high-grade obstruction carries a lifetime end-stage renal disease (ESRD) risk of up to **30%**.

	Lower urinary tract obstruction (LUTO)	Upper tract
Commonest	Posterior urethral valves (PUV) and urethral atresia ≈ 50%	Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1
Others	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21	Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, fibroepithelial polyp, stones, crossing vessel
Outcome driver	ESRD and pulmonary hypoplasia	Unilateral function loss; contralateral kidney buffers

Presentation depends on site, severity, age and duration: prenatal hydronephrosis (most at 18–20 wk); after birth, mass, fever or recurrent urinary tract infection (UTI), flank pain, polyuria or retention, failure to thrive, electrolyte imbalance, or respiratory failure from pulmonary hypoplasia.

> **KEY POINT**
> Trap: **detrusor hypertrophy** is a lower-tract cause of upper-tract (UVJ) obstruction, because the thick wall narrows the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

2 · Haemodynamic changes

Phased flow changes explain injury and diuresis.

Urine production continues, so pressure rises; net filtration pressure, glomerular filtration rate (GFR) and renal blood flow (RBF) fall, causing ischaemia. Injury begins within **2 h** of complete obstruction, worse if bilateral or solitary.

† Phase (complete)	Unilateral	Bilateral / solitary
I (0–1.5 h)	RBF rises, ureteric pressure rises (afferent dilatation: prostaglandin E2, nitric oxide)	Modest RBF rise
II (1.5–5 h)	RBF falls, pressure still high (efferent constriction)	RBF falls, pressure rises
III (> 5 h)	RBF and pressure fall (afferent constriction: angiotensin II, thromboxane)	Pressure stays high (retained solute, atrial natriuretic peptide) — basis of post-obstructive diuresis

- **Hypertension** is renin–angiotensin driven: bilateral ≈ **77%**, unilateral ≈ **20%** (renin lateralises).

3 · Tubular function

Tubular damage explains polyuria and salt wasting.

- **Aquaporins 1–4 and sodium transporters are downregulated**, causing a concentrating defect, salt wasting, polyuria and **distal renal tubular acidosis**.
- Early, the kidney wastes Na, K, HCO_3, Mg, Ca and phosphate; later it retains K, H^+, Cl and ammonium. So potassium is **wasted early, retained late**.
- Fractional excretion of sodium (FENa) is < 1% in acute obstruction and > 1% with tubular injury.

4 · Pathology and fibrosis

Fibrosis is the common end point.

- In the immature kidney, apoptosis and impaired nephrogenesis permanently cap nephron number; in the mature kidney, tubular proliferation leads to collagen and scarring.
- Fibrosis (raised α-smooth muscle actin and extracellular matrix) is the final common path. Stretch starts it; tumour necrosis factor-α (TNF-α) and interleukins (IL) activate **transforming growth factor-β (TGF-β) and Smad**, amplified by angiotensin II; **miR-34a** drives epithelial–mesenchymal transition.
- **Bladder**: hypertrophy, then compensation, then decompensation; IL-6, TNF-α and monocyte chemoattractant protein-1 reduce compliance. Storage pressure > **40 cm H_2O** injures the kidneys.

5 · Recovery and compensatory growth

Relief does not guarantee recovery.

- Relief can reverse damage, but **incomplete recovery is common**. Obstruction erodes reserve, so acute-on-chronic kidney disease (CKD) quickly becomes acute kidney injury (AKI).
- Compensatory growth (IGF-1, TGF-β1, HGF) is **glomerulomegaly, not new nephrons**.

6 · Post-obstructive diuresis (POD)

Expect it after bilateral relief; do not drive it with fluids.

- P1 Post-obstructive diuresis — *unilateral ≈ **2%**, bilateral ≈ **50%**; aquaporin/transporter loss, medullary washout, retained solute; risk after valve ablation (oligohydramnios, preterm, creatinine); monitoring, partial replacement, avoid overcorrection*

Not in the paper answer:

- Preterm infants have low aquaporin-2, hence POD risk.
- Bilateral cases lack a normal kidney to regulate volume.

> **ALGORITHM**
>
> 1. Monitor intake, output, and serum and urine electrolytes.
> 2. Replace losses and correct electrolytes.
> 3. **Do not over-replace**: it perpetuates POD.
> 4. Expect resolution within **24 h**; it seldom lasts > 48 h.

7 · Fetus and child

Fetal obstruction threatens the lungs as well as the kidneys.

- P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — *UTD APD thresholds (**≥ 7 mm** at 16–27 wk, **≥ 10 mm** ≥ 28 wk), serial fetal urine (Na < 100, Cl < 90, osmolality < 200, β2-microglobulin < 6), shunt/fetal cystoscopy, amnioinfusion*

Not in the paper answer:

- Kidneys are seen from 9 wk, the bladder by 12 wk (not seen by 15 wk is abnormal); a visible fetal ureter is dilated. Megacystis is a bladder failing to empty over 40 min at ≥ 18 wk.
- Second-trimester oligo- or anhydramnios carries **60–80%** mortality.
- Favourable fetal urine also has Ca < 8 mg/dL; the last sample reflects current function.
- Whitaker test: obstructed if pelvic pressure > **12 cm H_2O** or rises continuously. Cystatin C GFR is most accurate in children.
- After birth: ultrasound first; mercaptoacetyltriglycine (MAG3) for drainage; dimercaptosuccinic acid (DMSA) for scars; voiding cystourethrogram (VCUG) for outlet and reflux.

8 · Treatment principles

Temporise, then define.

Drain to buy function, reconstruct later, and manage POD and electrolytes carefully.

Option	Indication / key point
Clean intermittent catheterisation (CIC)	Neurogenic bladder; reduces UTI, stones, retention
Overnight catheter drainage	Valve bladder with nocturnal polyuria and progressive CKD
Vesicostomy	Infant with complex LUTO, unstable or unable to do CIC
Percutaneous nephrostomy	Upper tract; needs some hydronephrosis
Stent / cutaneous ureterostomy	Temporising; stent excellent for extrinsic obstruction
Pyeloplasty / reimplantation	Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children

- Use non-steroidal anti-inflammatory drugs only if function is preserved. For oliguria or AKI, treat the cause and optimise fluids and perfusion; in CKD, control blood pressure and diet. ESRD needs dialysis or transplant (*Ch 61*).

52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

1 · Definitions and classification

Both lesions are one defect; the definitions decide incontinence risk and which operation suits.

- Ectopic ureter and ureterocele are one defect of **ureterotrigonal connection** on a continuum, with intermediate cases. The appearance changes the technique, not the strategy.
- An **ectopic ureter** is a single or duplex ureter that does not enter the trigone. Ectopia means entry **at or distal to the bladder neck**; merely opening near it is not ectopic.
- A **ureterocele** is **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra. It is never entirely urethral and never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

	Ectopic ureter	Ureterocele
Distal ureter	Dilated or not, opens ectopically	Cystic dilation
Position	At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M)	Bladder or spanning bladder neck and urethra
Wolffian attachment	Yes (male)	Never
Incontinence (untreated)	Yes, in females	No

Ureterocele classification

- P3 Ureterocele — *Glassberg intravesical vs ectopic (most useful, predicts TUI outcome); Stephens incl. cecoureterocele; Churchill functional*

Not in the paper answer:

- Churchill groups run from upper pole only, to the whole ipsilateral kidney, to the contralateral kidney too (through reflux or bladder outlet obstruction, BOO); the last group is ~1/4. Its outcome prediction is unreliable.
- A **cecoureterocele** has a large incompetent intravesical orifice, and its cavity tunnels submucosally beyond the bladder neck into the urethra. It is often missed before surgery, troublesome at incision and a risk for incontinence.
- Say "single intravesical", not "simple", ureterocele: it can still obstruct severely.
- Distinguish an ectopic ureterocele from an intravesical one that **prolapses** into the urethra only on voiding.
- **Ureterocele disproportion** (non-obstructive): a small dysplastic, non-dilated upper pole invisible on imaging and no ureteric dilation, so a typical ureterocele sits with an apparently normal kidney.

2 · Embryology essentials

Bud position explains why the upper pole obstructs and the lower pole refluxes.

- In a duplex, the upper pole bud arises **later and more cephalad** on the wolffian duct. It is incorporated later, so its orifice is carried **caudally** and becomes ectopic.
- **Weigert–Meyer rule**: **the upper pole orifice lies caudal (and medial) to the lower pole orifice**. So the upper pole gets ectopia or ureterocele (obstruction), and the lower pole gets a short tunnel and vesicoureteral reflux (VUR).
- Failure of the **Chwalle membrane** to rupture causes a ureterocele (classic theory).

- Classically the common nephric duct forms the trigone. In the revised view it regresses by apoptosis, and the trigone comes from endoderm.
- Renal dysplasia lies **downstream** of the connection defect, not parallel to it.
- An ectopic ureter never enters müllerian structures directly. It reaches the vagina, cervix or uterus through the **Gartner duct**; a cystic Gartner duct rupturing into the vagina underlies female incontinence.

3 · Ectopic ureter — sites and sex asymmetry

The site relative to the external sphincter decides whether the child leaks.

- P3 Embryology and management of ectopic ureter. — *female vs male sites, continuous dribbling vs never incontinent, epididymitis, Zinner syndrome, Gartner duct*

Not in the paper answer:

- In Zinner syndrome, retrograde injection shows communication of ureter, seminal vesicle and vas.
- A **single-system ectopic ureter** may mimic an "absent kidney", because the small poorly functioning unit is missed on ultrasound; use contrast computed tomography (CT) or a renal function study.
- Distal obstruction may cause severe hydroureteronephrosis, sometimes enough to cause non-function.
- **Bilateral single-system ectopia** (rare) gives a hypoplastic bladder, poor trigone and bladder neck, and bilateral dysplasia.

> **KEY POINT**
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites are the bladder neck, prostatic urethra, vas, seminal vesicle and ejaculatory duct.

4 · Clinical presentation

Most are found prenatally; the rest present with infection, wetting, a prolapsing mass or late pain.

Prenatal

- Most are found as prenatal hydronephrosis, with the specific diagnosis made after birth. A duplex is hard to see unless one moiety is dilated. A fetal upper pole "cyst" is upper pole hydronephrosis until proven otherwise.
- **The diagnosis is made at the bladder**, because the upper tracts look identical. A large ureterocele can fill and mask the bladder.
- Record upper pole thickness and echogenicity. Dilation of the ipsilateral lower pole or contralateral kidney suggests reflux. Ureterocele BOO rarely causes oligohydramnios.
- Dilation of the whole kidney and ureter suggests a single system, which may mimic megaureter or severe reflux until after birth.

Incidental and infection

- A "presumed ovarian cyst" may be a dilated ureter, so include it in the differential of a cystic pelvic mass.
- Do ultrasound in every child with urosepsis, even after a normal late prenatal scan. If sepsis does not respond, do an **urgent transurethral incision (TUI)** of a ureterocele, or percutaneous or open drainage (usually end ureterostomy) of an ectopic ureter.

- An infected ectopic ureter gives subacute low-grade fever with spikes, and a **negative culture does not exclude infection**. Purulent perineal discharge in a girl needs ultrasound. An unobstructed ectopic ureter rarely becomes infected.
- **Any young male with bacterial epididymitis needs upper tract ultrasound**; it is often recurrent before the diagnosis is made.

Incontinence

- Evaluate a toilet-trained girl with verified **continuous low-volume dampness day and night** and normal voiding. Untreated ureteroceles do not cause incontinence.
- The key question is: **can she be dry for 30–60 minutes?** With an ectopic ureter she cannot, though parents of other wet children also say "always wet".
- Before toilet training, look for dribbling at nappy changes. Rarely the leak is intermittent, through a Gartner duct membrane.
- The leaking moiety may be non-dilated and missed on imaging, so suspicion rests on the history. Exclude voiding postponement, posturing and constipation.

Ureterocele prolapse

- It appears as a **smooth, congested, mucosa-covered interlabial mass** from the urethra, often with voiding difficulty. A lobulated grape-like mass suggests **sarcoma botryoides** instead.
- Bladder ultrasound confirms it. A voiding cystourethrogram (VCUG) shows the prolapse, lower pole reflux and the "drooping lily".

Late presentation

- Teenagers and adults present with infection or pain, rarely incontinence. An adult single-system non-obstructing ureterocele often contains a **stone**.
- The moiety usually has little function, so upper pole heminephrectomy is the mainstay; upper-to-lower ureteroureterostomy is the alternative.

5 · Evaluation

Imaging defines anatomy, reflux and moiety function; cystoscopy plans the incision.

- Examine the perineum of every continuously wet girl. The ectopic orifice lies **between the urethral meatus and the vagina**; cannulate it with an angiocatheter for contrast. A dilated upper pole may be palpable in an infant.

Imaging

Test	Answers	Note
Ultrasound	Anatomy; inferred function; bladder	First always; cannot separate ectopic ureter from ureterocele above bladder
Voiding cystourethrogram (VCUG)	Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra	Optional if obstruction alone dictates incision; repeat after incision
Dimercaptosuccinic acid (DMSA)	Differential function — gold standard	Only "some vs none"
Mercaptoacetyltriglycine (MAG3) renogram	Function and drainage	Observation candidates; after 2 months

Test	Answers	Note
MR urography	Most detailed anatomy and function	Complex anatomy, non-dilated upper pole; needs sedation
Intravenous pyelogram (IVP)	Anatomy and qualitative function	Non-dilated ectopic ureter with incontinence

- A **ureterocele** is a thin-walled cyst within the bladder. A **pseudoureterocele** is a dilated ectopic ureter indenting the bladder; its wall is **bilaminar and thicker, and its lumen lies outside the bladder**.
- Pitfalls: a large ureterocele fills the bladder; a full bladder effaces it so it mimics an ectopic ureter; a lobulated one looks like two cysts; cecoureterocele extension is hard to see. So watch through filling and voiding, which also shows emptying and wall thickness.
- On VCUG, lower pole reflux is recognised by calyceal orientation. The **"drooping lily"** is the lower pole pushed down and out by a dilated upper pole. With BOO or severe bilateral obstruction, puncture on ultrasound alone and do the VCUG after puncture.
- † On excretory urography a single-system intravesical ureterocele gives a **"cobra head" (spring-onion)** sign: a dilated distal ureter with a radiolucent halo.
- There are no criteria for "normal" upper pole function, and **preoperative function does not predict function after decompression**.

VCUG terms

Term	Meaning	Significance
Effacement	Flattens with filling	Mimics ectopic ureter
Prolapse	Moves into urethra	Outlet obstruction; predicts secondary surgery
Eversion	Diverticulum-like protrusion of bladder base	**Weak trigonal floor, so open repair likely**

Endoscopy

- **Choose the site** before incising. Start with little filling, because the lowest part of the ureterocele (the best incision site) is seen then.
- Document the urethra, bladder neck, trigone and orifices. Detect any urethral extension (cecoureterocele) so the incision leaves no obstructing segment.
- Retrograde injection confirms ureterocele disproportion or a ureter–vas–seminal vesicle communication.
- An ectopic orifice at the bladder neck is often patulous. An ectopic ureter separated from the urethra by a thin layer mimics a ureterocele and goes with bladder neck defects, so it needs wider excision with bladder neck reconstruction.
- A patulous bladder neck or cecoureterocele carries a later incontinence risk; consider bladder neck tightening (videourodynamics may help).

6 · Management principles

Decisions are empirical, because function before and after decompression is unpredictable.

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.
- Start daily antibiotic prophylaxis early in significant hydronephrosis, even without reflux.

- Relieving obstruction may **create** reflux, or may let lower pole reflux resolve. This is the central trade-off.
- Function is unpredictable before and after decompression, and recovery is limited, so decisions are empirical.
- **Fetal intervention**: none for ectopic ureter. For ureterocele, only if it is large and obstructs all units (aspiration or laser, experimental).

Neonatal algorithm

> **ALGORITHM**
> 1. After prenatal hydronephrosis, do postnatal ultrasound.
> 2. If there is significant hydroureteronephrosis, start prophylaxis.
> 3. Do VCUG (optional if it will not change the decision to decompress), then a renal scan after 2 months.
> 4. If the ureterocele is small, with minimal hydronephrosis and no obstruction, then observe.
> 5. If obstruction is significant and parenchyma is at risk, then do TUI (or a cutaneous distal ureterostomy if endoscopy is impossible, until old enough for reimplant).
> 6. If sepsis does not respond, then do TUI (ureterocele) or nephrostomy or end ureterostomy (ectopic ureter); elective drainage is still needed later.

- The best age for elective decompression is undefined.
- For an infant ectopic ureter, a side-to-side, non-dismembered, refluxing anastomosis to the bladder is meant to be definitive, but there are no long-term data.
- **Observation** suits selected ureteroceles: no obstruction of the ipsilateral lower pole or contralateral kidney and lower pole reflux ≤ grade III; a non-functioning upper pole (e.g. multicystic dysplastic, no ureteric dilation); or no obstruction on renogram. Dilation and reflux may resolve, but counsel about unpredictable acute presentation.

7 · Operative options

Each operation trades renal risk, bladder risk and the chance of a second operation.

Transurethral incision (TUI) of ureterocele

- TUI relieves obstruction in **78–97%**. It is the choice in **sepsis** and in the **infant**.
- Make a **low transverse full-thickness incision** with cutting current, **as distal and close to the bladder floor as possible**, to reduce reflux. Use an angled-tip wire, Collins knife or laser, and incise deeply. A urine jet or urothelium seen inside confirms adequacy.
- For an ectopic ureterocele, extend a longitudinal incision into the urethral part, or make separate intravesical and urethral punctures, so no obstructing urethral flap remains.
- The **"watering-can"** technique uses multiple holmium punctures; it decompresses equally with less de novo reflux.
- A well child needs no catheter and goes home the same day. Do ultrasound and VCUG at **4–6 weeks**.
- New reflux occurs in **0–50%**. **Intravesical** ureteroceles decompress without reflux in 70–80%; **extravesical** ones need secondary surgery in up to 70%. Reoperation risk ratio is 1.74 with lower pole reflux and 2.78 if extravesical (not additive). Single systems do better than duplex.

- Persisting reflux usually leads to ureterocele excision and reimplantation, though some resolves.

> **KEY POINT**
> Why TUI first even if half need more surgery? Outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. Less appealing in an older child with a massive non-functioning upper pole — definitive surgery at diagnosis.

Upper pole partial nephrectomy (heminephrectomy)

- It is for a non-functioning upper pole, or doubtful drainage from massive dilation. It can be open (flank or dorsal lumbotomy), laparoscopic or robotic. The main concern is **not to damage the viable lower pole**.
- Key steps: retract gently; use the upper pole ureter as a handle and pass it **behind the renal vessels**; ligate the upper pole vessels (usually 2–3); resect at the demarcation; close over a fat pedicle. Give mannitol if the pedicle is clamped and papaverine for vasospasm. Laparoscopically, ties avoid clip avulsion, and intravenous methylene blue checks for leaks.
- **Dissect on the wall of the upper pole ureter**, and leave its back wall on the lower pole ureter near the bladder, because they share a blood supply.
- A small distal stump is ligated and left. A large stump, or one that is obstructed and refluxing, is excised low without injuring the bladder neck or vas.
- Do postoperative Doppler ultrasound to look for urinoma and check lower pole flow. Clipping the ureter alone is tentative; pyonephrosis has been reported.
- Outcomes: for ectopic ureter it is usually curative. For ureterocele, lower pole reflux resolves in up to 20%, new reflux appears in **15–50%**, and secondary surgery is needed in **40–50%**.

Ureteroureterostomy / pyeloureterostomy

- The upper pole is joined end-to-side into the lower pole system. It is used for functioning and, increasingly, non-functioning upper poles; the feared hypertension has not been substantiated.
- For a **distal** (inguinal) repair, **stent the lower pole ureter cystoscopically first**. A **proximal** repair is preferred for a dilated upper pole or size mismatch.
- Keep medial dissection minimal and resect the upper pole ureter low on its wall. Stent the recipient only, never across the anastomosis. A well-decompressed ureterocele stump is not excised.
- Robotic-assisted ureteroureterostomy (RALUU) matches open surgery with a slightly shorter stay; size mismatch is no barrier.

Lower tract reconstruction

- **Ureterocele excision with common sheath reimplantation** relieves obstruction and reflux at once, and is the fallback for persistent reflux. It risks the bladder neck, sphincter and vagina.
- Mobilise **both ureters as a unit** and never separate them, because they share a blood supply. Taper the upper pole ureter if needed, excise the ureterocele to the bladder neck, **plicate the detrusor defect** and reimplant submucosally. Below the bladder neck, protect the sphincter and fulgurate any unexcised remnant.
- A cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve, so watch for voiding dysfunction.

- **Marsupialisation** (excising the roof without back-wall dissection) suits small ureteroceles without a detrusor defect; complete excision is preferred.
- Reflux persists in **5–10%**, more often after tapering.
- **Total reconstruction** (heminephrectomy, excision and lower pole reimplant) suits an **older child with a massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

End ureterostomy for ectopic ureter

- It is for a neonate with sepsis or massive dilation; it decompresses, and function is assessed later.
- **Do not trim the redundant ureter**, because it shortens and the length is needed for reimplant. Site the stoma at the lateral end of a future Pfannenstiel incision.
- Do DMSA at ~4 months or after 6 months. If the moiety is salvageable, reimplant or do a ureteroureterostomy; if not, do nephrectomy.

8 · Decision summary

Upper pole function and lower pole reflux pick the operation.

Ectopic ureter (duplex)

ALGORITHM

1. If the upper pole is worth keeping and the lower pole refluxes, then do a common sheath reimplant, or a lower pole reimplant with distal ureteroureterostomy.
2. If the upper pole is worth keeping and there is no lower pole reflux, then do a proximal or distal ureteroureterostomy.
3. If function is unclear, or there is massive dilation or sepsis, then do a temporary end ureterostomy, then DMSA, then reconstruct or remove.
4. If removal is chosen (marked dilation, preference), then do upper pole nephrectomy with ureterectomy.

- If lower tract surgery is needed anyway for reflux, upper tract surgery may be avoided.
- For a single system, preserve or remove according to function and preference; there are no outcome data either way.

Ureterocele (Table 52.1)

Procedure	Ideal indication	Advantage	Limitation
TUI	Infant; large ureterocele with VUR	Outpatient; effective; occasionally definitive	Reflux into ureterocele moiety needs bladder surgery
Upper pole nephrectomy	Older; large non-functioning upper pole; no VUR	Removes pathology; avoids bladder	Lower pole risk; may still need bladder surgery
Ureteroureterostomy / pyelostomy	Older; functioning upper pole; no VUR	Drains with little risk	Ureterocele left; may reflux
Excision and common sheath reimplant	VUR; functioning upper pole without major dilation	Removes obstruction and reflux; no renal risk	Complex; vagina/bladder neck risk; tapering

KEY POINT

VUR is the pivot: without VUR choose an upper tract solution; with VUR, a bladder-level solution, except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy vs ureteroureterostomy. TUI failure costs little and may simplify the next operation.

9 · Complications

The worst is loss of the healthy lower pole.

- **Loss of lower pole function** after heminephrectomy is the most significant and may be recognised late. Prevent it by preserving the inter-ureteric blood supply and using gentle traction.
- **Urinoma** occurs in up to 20% after laparoscopic or robotic cases, mostly when the defect is not closed; observe it if asymptomatic.
- **De novo reflux** after TUI or heminephrectomy is observed if there is no infection; otherwise do excision and reimplantation.
- **Voiding dysfunction**: infrequent voiding is common and incontinence uncommon. Poor trigonal support lets the trigone balloon behind the bladder neck, causing obstruction, poor emptying, infection and recurrent VUR (also after TUI of a cecoureterocele).
- Assess with VCUG **including lateral voiding views**, with or without videourodynamics. Treat by trigonal or bladder neck reconstruction, clean intermittent catheterisation (CIC) or bulking.

10 · Other ureteral anomalies

Duplication is common; the rest are rare but have classic exam features.

Duplication

- It occurs in ~**1 in 125** (0.8%) at autopsy, F:M 1.6:1, and is unilateral 6× more often than bilateral. **Look for contralateral duplication**: an occult second ectopic ureter explains persistent wetting.
- A **bifid pelvis** has a single ureter. A **bifid ureter** is a partial duplication joining below the ureteropelvic junction (UPJ).
- **The lower pole gets single-system disease** (UPJ obstruction, VUR); **the upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Manage lower pole UPJ obstruction as in a single system, with the option of pyeloureterostomy to the upper pole ureter. Upper pole UPJ obstruction is exceptionally rare.
- Lower pole VUR is often high grade, with reduced function and slower resolution. Options are common sheath reimplant, endoscopic injection or low ureteroureterostomy.

Triplication

- Smith types: 1, three separate ureters (**35%**, commonest); 2, incomplete with two orifices (21%); 3, trifid with one orifice (31%); 4, two ureters with three orifices (13%). Weigert–Meyer usually holds, but is violated in quadruplication.

Valves, diverticulum, polyp

- † **Ureteral valves** are transverse mucosal folds containing smooth muscle. Diagnosis needs proximal dilation, a normal ureter below and no other cause; excise the segment.

- † A ureteral **diverticulum** is usually a **blind-ending bifid ureter**; excise it if symptomatic.
- A **fibroepithelial polyp** is commonest at the **UPJ**, **89% male, 78% left**; only 22% are seen before surgery (as a filling defect). At the UPJ, do pyeloplasty; elsewhere, **remove it ureteroscopically**.

Vascular position anomalies

- P1 P2 IVC anomalous development and retrocaval ureter; Retroileal Ureter — *persistent subcardinal vein, right only, type I fishhook vs II, CT urography, transpose anterior; retroiliac at L5–S1 with mesonephric ectopia*

Not in the paper answer:

- A retrocaval ureter is always right-sided, even with bilateral cavae. It mimics UPJ obstruction, so a pyeloplasty fails.
- Ureteral herniation (inguinal, scrotal, femoral) is mostly paraperitoneal, with the ureteric loop **medial** to the sac.

53 Surgery of the Ureter in Children

1 · Pelviureteric junction (PUJ) obstruction — definition and aetiology

Dilatation is not obstruction; the cause (intrinsic, vessel or secondary) shapes the repair.

- P1 Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. — *Koff definition; intrinsic (infant) vs crossing vessel (older, intermittent pain) vs secondary; bilateral 10–40%; associated anomalies up to 50%; SFU grades; MAG3 T½ cut-offs; one-third need surgery*

Not in the paper answer:

- Paediatric hydronephrosis is a pelvic anteroposterior diameter (APD) of **12 mm**, but hydronephrosis is not obstruction. Koff's definition is retrospective, so every surgical indication tries to predict harm before function is lost.
- With a crossing vessel there is often no intrinsic narrowing after mobilisation, yet **dismembering and transposing the ureter anterior to the vessel prevents recurrence**.
- **Secondary PUJ obstruction** from high-grade vesicoureteral reflux (VUR): avoid reimplant and dismembered pyeloplasty at one sitting, because it devascularises the ureter from both ends. Manage conservatively or in **two stages**.
- **Lower-moiety PUJ obstruction in a duplex**: with incomplete duplication, do ureteropyelostomy to the lower-pole pelvis; with complete duplication, a standard pyeloplasty. If the moiety has ≤ **5–10%** function, do partial nephrectomy.
- Associations, in order of frequency: contralateral PUJ obstruction (commonest), dysplasia, multicystic dysplastic kidney (MCDK), then agenesis. VUR occurs in up to 40% and usually resolves. Horseshoe kidney and VATER also occur.

2 · Diagnosis and indications

Decide on function trend and dilatation, not on washout alone.

- Most are found on antenatal ultrasound (US) at ≈ 20 weeks. A febrile urinary tract infection (UTI) usually responds to antibiotics; few need nephrostomy.
- Neonatal renogram washout is confounded by immaturity, hydration, diuretic timing, posture and pelvic compliance, with false positives up to **44%**. So decide on the trend in differential renal function (DRF) and on dilatation, not on half-time (T½) alone.
- **Pyeloplasty indications**: increasing APD; low or falling DRF; breakthrough infection on prophylaxis; symptoms (pain; failure to thrive or feeding difficulty in infants; haematuria; rarely renin-mediated hypertension).

3 · Anderson–Hynes dismembered pyeloplasty

One principle, applied by any route: dismember, excise, spatulate and join without tension.

It is the **gold standard** for any access, with success ≈ **95%**; it was first devised for retrocaval ureter. It preserves a crossing vessel (the ureter is transposed in front of it), excises the adynamic segment and allows pelvic reduction.

ALGORITHM

1. Expose the PUJ and free the lower pole, looking for an accessory vessel.
2. Use bipolar diathermy sparingly and place stay sutures.
3. Decompress the pelvis with a 21-G needle, then dismember, with or without pelvic reduction.
4. **Spatulate laterally** into healthy ureter, keeping the stenotic segment as a "no-touch" handle.
5. Bring the apex of the spatulation to the inferior pelvis and make a tension-free, watertight 6-0/5-0 absorbable anastomosis.
6. Excise the stenotic segment last, irrigate out clot and cover with perinephric fat.

- For a long gap, mobilise the whole kidney and use a horizontal lower-pelvis incision. Excess stay-suture tension kinks the ureter when released. A drain is usually unnecessary.

Alternative	Best indication	Point
Foley Y-V-plasty	High insertion, **no vessel**	V base on dependent medial pelvis, apex at PUJ
Spiral flap	Long proximal stricture, dependent PUJ	Length:width ≤ 3:1
Vertical flap	Dependent PUJ, large box-shaped extrarenal pelvis	Flap length must match gap
Vascular hitch (Hellström)	Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery	Vessel fixed cranially in pelvic wall; no hypertension seen
Uretero-calicostomy	Failed repair, short scarred ureter, dilated lower calyx with thin cortex	Lower-pole nephrectomy exposes calyx; spatulated ureter joined over stent; capsule not closed near anastomosis

Use flaps only for a long stenosis where a tension-free primary anastomosis is impossible. Non-dismembered (Heineke–Mikulicz) repair is inferior.

KEY POINT

Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

KEY POINT

One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Access, stent and robot are choices of convenience; failure is managed by repeating the principle.

4 · Access

Open in infants; laparoscopic and robotic results now match open.

- In **infants**, open surgery is preferred. **Posterior lumbotomy** suits only infants and toddlers; the flank approach (11th–12th rib tip) suits any age.
- Laparoscopic results equal open (meta-analysis). Robotic surgery has a shorter learning curve and faster resolution of hydronephrosis.

	Transperitoneal	Retroperitoneal
For	Easier landmarks; left transmesocolic	No bowel injury; leak extraperitoneal; shorter stay
Against	Bowel risk; urinary ascites causing ileus	Learning curve; small space
Note	Preferred for horseshoe kidney, retrocaval ureter	One randomised trial favoured it; 80–90° flank, table **not flexed**, balloon 200–300 mL

- **Endopyelotomy** or balloon dilatation does poorly long-term under 18 years. Its niche is **failed pyeloplasty** without a vessel, with a double-J stent for 6 weeks.
- Stones found at pyeloplasty are removed with a flexible scope through a port.

5 · Stents, outcomes, failure

Stent choice is convenience; failure is treated by repeating the principle.

Stent	Removal	Point
Internal double-J	Second anaesthetic unless string/magnet	Catheter 24–48 h; string gives more febrile UTI (girls)
Externalised nephro-ureteric	7–14 days, no anaesthesia	Fewer complications, shorter stay in infants
Stentless	—	Needs flawless technique

- Stent duration of 1–4 weeks does not change outcome. An antegrade double-J may not pass the infant ureterovesical junction (UVJ); check with methylene blue in the bladder.
- For emergency diversion before repair, use a **nephrostomy, not a retrograde stent**, because a stent inflames the PUJ.
- Success is 90–100%, though variably defined. The commonest reason for conversion is bleeding.
- P1 Management of failed pyeloplasty — *early failure = oedema/clot: wait with or without nephrostomy; late: endopyelotomy or redo transperitoneal pyeloplasty (best); ureterocalicostomy for short scarred ureter*

Not in the paper answer:

- The strictest definition of success is no reintervention.
- After late failure, balloon dilatation or endopyelotomy needs a stent for 4–6 weeks.

6 · Primary megaureter

Most resolve; the challenge is spotting the few that are truly obstructed.

- P3 Primary obst. megaureter. — *≥ 7 mm; four types; adynamic juxtavesical segment maturing craniocaudally; < 10 mm rarely needs surgery; indications incl. DRF < 40% or fall ≥ 5%; temporising options; excise–taper–5:1 reimplant; Starr/Kalicinski/Hendren; success > 90%*

Not in the paper answer:

- The refined Hendren excises only mucosa and muscle, sparing the adventitia. Use a locking suture on the proximal two-thirds and interrupted sutures distally, and taper gradually, because an abrupt calibre change obstructs.
- Primary obstructive megaureter (POM) does better than refluxing megaureter (stiff, collagen-rich UVJ), and worse with valves or neurogenic bladder.
- Transureteroureterostomy is an option for unilateral severe scarring.

- The hard part is separating obstructed from non-obstructed non-refluxing ureters, because dilatation does not prove obstruction. Follow into adulthood, since some worsen at puberty.
- In infancy, the small bladder cannot take a safe non-refluxing tunnel for a massive ureter, so temporise. Options: **distal loop cutaneous ureterostomy** (inguinal; stomal stenosis rare); a **refluxing reimplant** (the "lesser evil"); balloon dilatation until the waist disappears (weak, retrospective evidence; migration, infection, VUR and stones); nephrostomy, which is hard to maintain.

> **KEY POINT**
> Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity, so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

> **ALGORITHM**
> 1. Excise the stenotic distal segment and straighten the ureter.
> 2. Taper if needed.
> 3. Reimplant non-refluxing at a **5:1** tunnel length:diameter.
> 4. Stent with a feeding tube for 7 days or a double-J for 4 weeks.

- Plication and folding preserve the blood supply. Hendren tapering (excising the lateral wall over an 8–10 Fr catheter) risks ischaemic stricture.
- For a very large ureter, an extravesical approach avoids a huge hiatus. Laparoscopic and robotic results are comparable.
- Complications: treat obstruction with a stent first; an ischaemic stricture needs redo. Reflux is observed or injected, then revised.

7 · Surgery for vesicoureteral reflux

The choice of reimplant trades tunnel length against kinking, bladder morbidity and nerve injury.

> **KEY POINT**
> **Principles**: exclude/treat bladder causes first · handle ureter (blood supply) and bladder (haematuria, spasm) gently · tension-free mobilisation · tunnel length:diameter **5:1** (Paquin) · straight course at empty and full bladder.

- Techniques are classified by access, by approach (intravesical or extravesical) and by tunnel position relative to the original hiatus (suprahiatal or infrahiatal). Use a Pfannenstiel incision, or a vertical midline for redo.

Technique	Approach / tunnel	Strength	Weakness
Politano–Leadbetter	Intravesical; suprahiatal neo-hiatus, vertical tunnel	Long tunnel for high grade	Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior
Glenn–Anderson	Intravesical; infrahiatal toward bladder neck	Least kinking	Short tunnel
Cohen	Intravesical; cross-trigonal	Commonest; success > **98%**; thick bladder; with bladder-neck reconstruction	Difficult later retrograde catheterisation

Technique	Approach / tunnel	Strength	Weakness
Lich–Gregoir	Extravesical detrusorotomy ≈ 5 cm	No haematuria/spasm; short catheter	Transient voiding dysfunction/retention up to **10%**, esp. bilateral
Robotic extravesical (RALUR-EV)	Y-shaped detrusorotomy 4–5 cm	Less pain, spasm, stay	Early series: leak, obstruction, ureteric injury

Cohen — key steps

ALGORITHM

1. Open the bladder vertically and put a feeding tube in the orifice as a handle.
2. Circumscribe the orifice and free the ureter, preserving its distal blood supply, until it reaches the contralateral wall without tension.
3. Make a submucosal tunnel across the trigone, with the new orifice above the contralateral one.
4. Suture mucosa to mucosa with 6-0, with a detrusor hitch at 6 o'clock.

For later access to a cross-trigonal orifice, use a flexible ureteroscope or a suprapubic needle aimed at the orifice.

Lich–Gregoir — key steps

ALGORITHM

1. With the bladder one-third full, divide the obliterated umbilical artery.
2. Free the ureter **hugging it**, only as far as the UVJ.
3. Distend the bladder and mark a 5-cm tunnel; incise the detrusor down to mucosa and raise wide flaps.
4. Lay the ureter in and close the detrusor caudocranially, catching adventitia at the apex.
5. Leave a Foley catheter for 12–48 h.

KEY POINT

Why voiding dysfunction? The pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ. Stay on the ureter, avoid the dorsomedial UVJ, stop at the UVJ. The Zaontz modification (detrusorotomy around the UVJ) is now avoided.

- In the American Urological Association (AUA) meta-analysis, open success is **95.9%**; minimally invasive results are converging.
- **Follow-up**: US at 6–12 weeks. A voiding cystourethrogram (VCUG) may be omitted after low-grade reflux if US is normal and there is no UTI or dysfunction. Do it for high grade, new hydronephrosis, UTI or a dysfunctional bladder.

Complications of reimplantation

	Early	Long-term
Reflux	Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves, so observe; intervene for pyelonephritis. Routine bilateral repair not justified	Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder): treat the bladder first (anticholinergic and clean intermittent catheterisation)

	Early	Long-term
Obstruction	Oedema, clot, haematoma, kink within 2 weeks: double-J or nephrostomy; most resolve	Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis: balloon and stent, then re-reimplant

- Early mild hydronephrosis should not exceed the maximum dilatation that the reflux caused.
- Contralateral reflux risk rises with age < 6 years, capacity < 50% of expected and a refluxing duplex; technique is irrelevant.
- **Reoperation**: excise ischaemic ureter back to bleeding, peristaltic tissue. For a short ureter use a **psoas hitch** (one side only), Boari flap, transureteroureterostomy, nipple valve or a Monti bowel ureter.

> **KEY POINT**
> Early complications are tissue reactions — drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder; reimplanting into an abnormal bladder worsens dilatation.

Endoscopic injection

- P3 Endoscopic correction of VUR (STING procedure): indications and complications — *Dx/HA standard, ≈ 90% low grade; mound within 0.1–0.2 mL, volcano; never cauterise; STING vs HIT vs double HIT; late obstruction (Vantris); Swedish trial 20% recurrence at 2 years*

Not in the paper answer:

- Abandon the injection if the bladder is inflamed.
- After failed injection, reimplantation is not harder, because the Deflux implant is encapsulated.
- Particles < 50 µm reach the systemic circulation and those < 80 µm are phagocytosed, which is why Teflon migrated.
- After failure, repeat the injection once or twice, then reimplant.

8 · Other ureteric lesions

Rare mimics of commoner lesions.

- † **Retrocaval ureter** is right-sided and passes behind the inferior vena cava (IVC; persistent posterior cardinal vein), with a fish-hook or S shape. Re-anastomose it in front of the cava.
- A **congenital mid-ureteric stricture** is often misread as megaureter. Excise it (1–3 cm) and do a ureteroureterostomy.
- A **fibroepithelial polyp** occurs in the upper third, in males, on the left, with pain and haematuria. Confirm by retrograde study or ureteroscopy, and remove it by endoscopic polypectomy or resection.
- **Retroperitoneal fibrosis**: give steroids first, then azathioprine, then a double-J stent, and finally ureterolysis.

56 Hypospadias

1 · Definition, diagnosis and embryology

Arrested urethral closure explains meatus, hood and curvature.

- The **triad** is a proximal (ventral) ectopic meatus, ventral curvature and a dorsally hooded foreskin (curvature and hood not always present). Incidence is ≈ **1 in 150–300** male births; diagnosis is clinical.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%, hides a wide distal meatus under a normal foreskin, so retract before every newborn circumcision.
- Avoid circumcision with curvature or ventral skin deficiency, as the foreskin may be needed.
- P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — *SRY, testosterone, 5α-reductase type 2, DHT at 8–12 weeks; endodermal two-zipper vs outdated Glenister ectodermal theory; dorsal prepuce fusion first*

Not in the paper answer:

- Ventral prepuce fusion awaits urethral fold fusion (≈ 20 weeks), so an ectopic meatus has an incomplete foreskin in **95%**.
- Glenister's older theory: the glanular urethra forms by ectodermal ingrowth fusing at the corona.
- The critical window is **weeks 8–16**, and closure runs from the anus to the glans tip. The two-zipper theory is Kurzrock's (current). MCQ

> **KEY POINT**
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close ventrally over an open urethra, and unfused dysplastic ventral tissue tethers the shaft into curvature.

2 · Aetiology

***Multifactorial**, but all causes converge on too little androgen action in weeks 8–16.*

- **Fetal and placental** factors: prematurity, small for gestational age, growth restriction and placental insufficiency (≈ 2× risk); assisted reproduction is an independent risk.
- **Genetic**: twin concordance 9–27%; first-degree relative 9–15%; chromosomal abnormality up to 30%. No single gene is responsible (HOX, FGF; WT1, SOX9, DMRT1, GATA4; LH receptor, 5α-reductase, androgen receptor). About 200 syndromes include it (WAGR, Denys–Drash, Smith–Lemli–Opitz).
- **Endocrine disruptors** (pesticides, oestrogens/progestins, diethylstilbestrol, phthalates, phytoestrogens) are proven in animals but unproven in humans.
- **Testicular dysgenesis syndrome**: hypospadias, cryptorchidism, poor semen and testis cancer share one fetal exposure.
- Shorter **anogenital distance (AGD)** means more severe hypospadias.

3 · Classification and severity

Grade severity in theatre, after degloving.

- Distal forms (glanular, coronal, distal shaft) are **70–85%**; proximal forms 10–25%.
- The meatus is the **least reliable** marker: **classify after degloving** by meatus, curvature and plate.

- P4 Hypospadias: present concepts in management, and hypospadias fistula — *GMS score table (glans, meatus, shaft; total 3–12, higher means more complications); POST low ratio favours inlay graft over plain TIP*

Not in the paper answer:

- The glans–meatus–shaft (GMS) score is Merriman's (2013) and is scored in theatre. A coronal meatus with a small flat glans and 45° curvature scores worse than its position suggests.

4 · Associated anomalies and DSD work-up

Proximal hypospadias with undescended testis may be a sex-development disorder.

- Hernia or hydrocele: 9–16%; undescended testis (UDT): ≈ 7%, ≈ 10% when proximal.
- **Disorder of sex development (DSD) work-up** is needed for UDT with proximal hypospadias, a perineal meatus or atypical genitalia. Since 2006 severe hypospadias is classed as 46,XY DSD. DSD risk is ≈ **50%** with a non-palpable vs 15% with a palpable testis, and 64% proximal vs 7% distal.
- Do ultrasound (US) for Müllerian remnants, karyotype, testosterone, follicle-stimulating hormone (FSH), luteinising hormone (LH) and anti-Müllerian hormone (AMH). 46,XY does not exclude 5α-reductase deficiency, androgen insensitivity, testicular dysgenesis or ovotesticular DSD (*Ch 59*).
- A **prostatic utricle** is commoner in proximal cases. A large one causes stasis, urinary tract infection (UTI) and difficult catheterisation; excision is rare (vas at risk).
- No routine voiding cystourethrogram (VCUG) or renal US.

> **KEY POINT**
> Hypospadias with cryptorchidism signals a systemic androgen problem, not a local one — hence karyotype before surgery.

5 · Pre-operative assessment

Set goals, timing and glans preparation.

- **Goals**: straight erection, urethra to the tip, acceptable cosmesis, fewest procedures, durable result. A straight penis voiding standing may gain little.
- Examine stretched length (exclude micropenis), groins, AGD, penoscrotal transposition, torsion, scrotal cleft and meatal stenosis; a "laser-beam" stream may justify surgery even when distal.
- **Timing**: genital awareness starts ≈ 18 months. The American Academy of Pediatrics (AAP, 1996) advises 6–12 months; the authors **6–18 months**. A systematic review (>16,000) found fewer complications when younger. One short anaesthetic is low-risk.
- **Androgen**: glans width **<14 mm** means more complications. Give testosterone cypionate **2 mg/kg intramuscularly at 5 and 2 weeks** before surgery if the glans is <15 mm (≈ 4 mm gain); dihydrotestosterone (DHT) or human chorionic gonadotropin are alternatives. Two randomised trials showed fewer complications; meta-analyses are inconclusive. Side effects (aggression, erections, pigmentation) resolve within 6 months.

6 · Intra-operative principles

Blood supply and tension-free closure prevent complications.

- Use a caudal (fistula link disputed), penile or pudendal block, one dose of ketorolac 0.5 mg/kg and intravenous antibiotics, with a tourniquet (released every 30 min) or epinephrine.
- Cystoscopy is only for severe proximal or redo cases.
- Rapidly absorbed sutures (chromic, polyglactin) beat polydioxanone; use 6-0–8-0. Slow sutures leave epithelialised tracks that become fistulae.
- Use magnification, **tension-free** closure, non-overlapping suture lines and haemostasis. **Never use hair-bearing skin** (stones, UTI). Higher surgeon volume means fewer complications.

7 · Curvature

Curvature decides one-stage or staged repair.

Causes: ventral skin tethering, dysplastic dartos and Buck's fascia, short urethra, corporal disproportion.

ALGORITHM

1. Deglove to the penoscrotal junction and excise dysplastic dartos.
2. Make an **artificial erection** (Gittes–McLaughlin, corporal compression or prostaglandin).
3. If straight, then proceed to urethroplasty.
4. If residual curvature is <30°, then do dorsal plication.
5. If >30°, then divide the urethral plate.
6. If still >30°, then do ventral corporal lengthening (usually staged).

- Mild is <30°, moderate 30–45°, severe ≥45°. Eyeballing underestimates, and a tourniquet may hide basal curvature.
- Plication (Nesbit, Baskin midline 12 o'clock, Heineke–Mikulicz) suits ≤30° (authors ≤15°); it risks nerve injury off the midline, pubertal recurrence and shortening.
- Lengthening uses transverse ungrafted corporotomies (staged tubularised autograft, STAG) or a single 9-to-3 corporotomy with graft (dermis 20% larger, tunica vaginalis, small intestinal submucosa). Risks are graft contraction and limited ventral skin.
- Recurrence is 28% after plication vs 9% after grafting, and not dividing the plate is an independent risk. Correct fully the first time.

8 · Urethroplasty

Match technique to plate and severity.

Order: curvature, urethroplasty, glansplasty, then skin with or without preputioplasty. Tubularised incised plate (TIP) repair is used for >90% of distal repairs.

Technique	Indication	Key point / pitfall
MAGPI (meatal advancement and glanuloplasty; Duckett 1981)	Glanular, mobile meatus, no curvature	Used more proximally gives a retrusive meatus
Urethromeatoplasty	Stenotic distal meatus	Heineke–Mikulicz
GAP (glans approximation)	Wide deep groove (MIP)	Two-layer neourethra
Thiersch–Duplay	Wide plate	Tubularise over 12 Fr
TIP / Snodgrass (1994)	Distal, midshaft; narrow plate; mild curvature	Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate

Technique	Indication	Key point / pitfall
Mathieu (flip-flap)	Distal, shallow plate	Narrow base causes ischaemia, fistula; fewer stenoses than TIP
Onlay island flap	Plate kept but narrow	Diverticulum if too wide; less stenosis
Duckett tube (transverse preputial island)	Proximal; plate divided	Higher complications; diverticulum
Two-stage Bracka	Proximal severe curvature, redo, poor plate	Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise
Two-stage Byars	Proximal severe curvature	Dorsal prepuce flaps; highest staged complications

- **STAG** has the lowest proximal complication rate (≈ 23%). Alternatives are STAC (three-stage, the authors' choice) and distal-first (Ulaanbaatar).
- Stage because urethroplasty over a fresh corporotomy or graft fails; repeat the artificial erection at stage 2. Proximal TIP is abandoned (devascularisation, recurrent curvature).

> **KEY POINT**
> Why proximal repairs fail — Poiseuille: resistance is proportional to length / radius⁴. A long, rigid, spongiosum-free tube ending in a non-distensible glans raises proximal pressure, causing fistula, diverticulum and stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty needs deep wings and a tension-free **two-layer closure**. A **barrier layer** (Belman dorsal dartos, ventral dartos, **tunica vaginalis** mobilised to the external ring, or spongiosum) reduces fistula, but must not compress the glans.

9 · Post-operative care

Dressing, diversion, spasm control.

- Leave a compressive dressing ≈ **72 h**. Divert urine **3–21 days** by complexity (dripping stent; suprapubic for older or redo); short distal repairs need no stent.
- Give oxybutynin until the stent is out. Stiff stents or lateral Foley pull cause glans dehiscence. Use paracetamol and ibuprofen; antibiotic prophylaxis evidence is mixed.

10 · Complications

Most complications trace to obstruction, ischaemia or tension.

Complication	Standard definition
Fistula	Urethral leak below meatus
Glans dehiscence	Wing separation; glans fusion <2 mm
Meatal stenosis	Obstructive symptoms and **<8 Fr** prepubertal / <12 Fr postpubertal
Diverticulum	Segmental sacculation on voiding
Recurrent curvature	**>30°** on erection

- P4 Hypospadias: present concepts in management, and hypospadias fistula — *commonest complication (<10%); distal obstruction key; wait 6–12 months; calibrate, stress test, layered closure with barrier; coronal fistula needs redo urethroplasty; recurrence ≈ 20%*

Not in the paper answer:

- An inlay graft avoids glans tension and so helps prevent dehiscence.
- Recurrent curvature comes from under-correction, graft contraction, skin tethering or pubertal disproportion.
- Hairy urethra causes narrowing, stones, UTI.
- **Meatal stenosis**: dilate, with or without steroid, then meatotomy or redo; commoner after TIP than Mathieu.
- **Glans dehiscence**: reoperate only for voiding symptoms.
- **Stricture** appears late; dilatation or urethrotomy rarely lasts, so do buccal urethroplasty, often staged.
- **Diverticulum**: excise the redundancy, keep the dartos and exclude distal obstruction.
- **Balanitis xerotica obliterans (BXO) / lichen sclerosus**: steroids usually fail, so excise all of it and do a staged buccal or preputial graft. Recurrence is up to **40%**, and it can invade buccal grafts.
- **Recurrent curvature**: deglove and test with artificial erection; do staged lengthening, or the Cecil procedure if skin is short. Treat a **hairy urethra** by laser epilation or replace it with a buccal graft.

> **KEY POINT**
> Nearly every complication traces to distal obstruction, ischaemia or tension — so fistula/diverticulum repair begins by calibrating the distal urethra.

Reoperation

- Devine: avoid "small operations"; excise scar widely and transfer tissue. Always do cystoscopy and artificial erection.
- **Buccal mucosa** is the graft of choice, with **lip mucosa** within the glans; bladder mucosa is abandoned (prolapse). Make the graft 10% larger and quilt it; do the urethroplasty ≥6 months later. In redo cases a dorsal inlay beats an onlay or tube.
- **Cecil–Culp**: bury the penis in the scrotum and release it at 9–12 months. Nitroglycerin, hyperbaric oxygen and tissue expanders are unproven.

11 · Outcomes and follow-up

Complications appear late; follow through puberty.

- A TIP meta-analysis (Pfistermuller 2015, 4,675 boys): complications 10.6%, reoperation 4.5%; fistula commoner than stenosis, then stricture; dartos helps. Distal success is 83–95%; proximal complications are 23–68%.
- About 50% of complications occur in year 1 and ≈ 75% by 3 years, so follow proximal cases **through puberty**. Recurrent UTI (≈ 2%) needs imaging for diverticulum, stricture or utricle.
- Outcome tools: HOSE (surgeon), Penile Perception Score (patient/parent), HOPE (photos, cosmetic).
- In adults, untreated mild distal hypospadias causes few problems. Proximal cases have worse satisfaction, sensation, fertility and ejaculation; penile length is the chief complaint; no major psychopathology. Even 20–30° of curvature can impair intercourse.

57 Etiology, Diagnosis and Management of the Undescended Testis

1 · Definitions

Why this matters: a retractile testis is watched; an undescended one is operated on.

- P1 P3 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis; Maldescended (undescended) testes and treatment algorithm — *undescended vs ectopic (SIP commonest), retractile, acquired (8–11 years), vanishing (monorchia/anorchia) vs agenesis*

Not in the paper answer:

- Rarer ectopic sites include perirenal and transverse (crossed) scrotal.

Term	Key point (beyond the papers)
Undescended testis (UDT)	Not in the normal scrotal position (midpoint at or below mid-scrotum). It is palpable or non-palpable; non-palpable means undescended or absent.
High scrotal testis	Stable, above the scrotal midpoint. A mixed group (stable, retractile, "gliding"), not routinely called undescended.
Retractile	One that rarely stays down or rests high may progress to acquired UDT.
Recurrent	Undescended at birth, descended spontaneously, then re-ascended (10–22%, at 1–5 years).
Secondary / testicular retraction	Suprascrotal after hernia repair / after orchidopexy (iatrogenic).
Agenesis	Never formed; comes with ipsilateral Müllerian persistence (extremely rare).

2 · Embryology of descent

Why this matters: the hormone driving each phase explains where a testis stops.

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *SRY/SF1, then SOX9, then Sertoli; INSL3 via RXFP2 + androgen; gubernaculum attaches to epididymis; transabdominal 10–15 weeks (outgrowth, CSL regression); scrotum by DHT at 10–12 weeks*

Not in the paper answer:

- The gubernaculum does not pull the testis; it swells, then regresses.
- Minipuberty (below).
- Fetal Leydig cells are active from 6–7 weeks and involute after 24 weeks.
- Insulin-like peptide 3 (INSL3) peaks at 15–17 weeks. Testosterone peaks at 14–16 weeks, driven first by placental human chorionic gonadotrophin (hCG), then by fetal luteinising hormone (LH).
- Anti-Müllerian hormone (AMH) regresses the Müllerian ducts at 9–10 weeks.
- Sertoli cells lack the androgen receptor early; otherwise AMH would be suppressed and spermatogenesis would start prematurely.
- Humans have no cranial gonadal ligament (unlike rodents).

Phase	Timing	Event	Control
Transabdominal (relative)	10–15 weeks	Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks	INSL3 and androgen act together

Phase	Timing	Event	Control
Inguinoscrotal	**20–28 weeks**	The canal widens and the processus forms; rapid passage through the canal (rarely before 22 weeks); the gubernaculum then regresses. Most are scrotal after 27 weeks.	Androgen via the genitofemoral nerve (**GFN**), which releases calcitonin gene-related peptide (**CGRP**); cremaster myogenesis (Wnt)

KEY POINT

Outgrowth creates the canal; regression lets the testis follow — a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

Minipuberty is a hypothalamic–pituitary–gonadal surge peaking at 1–3 months (earlier and stronger in preterm boys). LH, follicle-stimulating hormone (FSH), testosterone, AMH and inhibin B (INHB) rise. Sertoli cells proliferate, which sets final size, and gonocytes become **adult dark (Ad) spermatogonia**, the stem-cell pool. Failed transformation leads to apoptosis or malignant change. This is where a positional defect becomes a germ cell deficit.

3 · Epidemiology, aetiology and associations

Why this matters: spontaneous descent ends by about 6 months, which sets the waiting period.

- Prevalence is **2–4% at birth** (term), falling to **1–2% at 3 months**; up to 45% in preterm boys.
- Spontaneous descent occurs mainly in the first 3–6 months, rarely after 6 months. About 88% of high scrotal testes descend, but only about half of extrascrotal ones.
- 80–85% are isolated. Risk factors are **prematurity/low birth weight**, family history (twins > brothers > sons; maternal half-brothers > paternal), maternal diabetes and smoking.
- Diethylstilbestrol (DES) doubles risk; first-trimester corticosteroids in assisted reproduction add risk. Endocrine disruptors act as mixtures; none is proven singly.
- It is a polygenic threshold trait; rare INSL3, RXFP2 and NR5A1 variants occur.
- **Testicular dysgenesis syndrome** links cryptorchidism, hypospadias, infertility and testicular germ cell tumour (TGCT). A short anogenital distance marks low prenatal androgen; low placental weight is also associated.
- Non-syndromic UDT reflects primary gonadal dysfunction, not central hypogonadism: the LH/testosterone ratio is raised, and AMH/INHB are low if bilateral.
- No hormone predicts germ cell number, so hormones are not tested routinely outside disorders of sex development (DSD) and anorchia.

Associated condition	Cryptorchidism	Implication
Prune-belly syndrome	All (bilateral abdominal)	Abdominal orchidopexy
Spigelian hernia	53–80%	Absent gubernaculum/canal
Cerebral palsy	25–54%	Often bilateral
Gastroschisis	5–25%	**50–55% descend spontaneously — observe**
Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves	12–33%	Caudal-field anomalies are ipsilateral in 90%

Associated condition	Cryptorchidism	Implication
Congenital diaphragmatic hernia	7–30%	Always ipsilateral
Klinefelter (47,XXY)	> 50% of prepubertal Klinefelter boys	< 2% of non-syndromic UDT; more with hypospadias
Persistent Müllerian duct syndrome (PMDS)	Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22%	AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk

Congenital vs acquired

Feature	Congenital	Acquired
Diagnosis	Birth/neonatal	Mean 8–11 years
Position	Often higher	Lower; often superficial inguinal pouch (SIP)
Processus vaginalis	Patent (87% unilateral)	Usually closed (patent 45–50%)
Epididymal anomalies	37–60%	11–31%

An acquired UDT is a never-fully-descended testis unmasked by growth; it has fewer Ad spermatogonia and semen results like congenital cases. Retractile testes share this pathology and ascend in about 2–3% prospectively (more in referral series), so they need **annual examination through childhood**. Screening (Japan) reduces late surgery.

4 · Histology, associated anomalies and fertility

Why this matters: germ cells are lost from late infancy, which is the case for early surgery.

- Germ cell loss begins **after infancy** and progresses; the contralateral testis is affected less.
- The UDT is smaller at birth and grows less even after spontaneous descent: an intrinsic defect, worsened by position.
- Spermatogonia per tubule (S/T) is 2.5 at birth, with a nadir of 1.2 at 2–3 years; an age-specific reference is needed.
- Gonocytes disappear late and Ad spermatogonia form late (Ad absent in 7–85%); primary spermatocytes (normally at 4–5 years) are delayed. Sertoli numbers fall after 4 months, and type IV collagen is reduced in both testes.
- Pathology is similar whatever the position; germ cells are more with younger age, a palpable testis and less fibrosis.
- A **patent processus** is present in 87% of unilateral and 71% of bilateral cases; aberrant lateral gubernacular attachment in 66–75%.
- Epididymal non-fusion is commoner than elongation/looping, which is commoner than atresia; these track severity. A UDT is at higher torsion risk, which can mimic an incarcerated hernia.

Anomaly	Key features	Management
Polyorchidism	Usually left, scrotal in 75%; Kumar classification by drainage (A with vas, B without)	Observe a normal scrotal testis; orchidopexy if undescended with intact drainage
Splenogonadal fusion	Left side; the continuous form comes with limb defects and micrognathia	Recognise it to avoid a needless orchidectomy
Transverse testicular ectopia	Hernia with a contralateral non-palpable testis; PMDS in 20–50%	Laparoscopy; trans-septal placement if vasa fused

Trial	Design	Result	Take-home
Kollin et al.	Randomised trial: orchidopexy at 9 months vs 3 years	Early: more Sertoli and germ cells, better testicular growth (to 4 years)	Operate in first year; age at surgery, not hormones, predicts germ cells

Fertility outcome

Outcome	Former unilateral	Former bilateral
Oligospermia / azoospermia	31% / 14%	42% / 31%
Paternity	**≈ 90% of those attempting — as controls**	≈ 65% of those attempting (≈ 1/3 overall)

- Cryptorchidism accounts for 10–17% of men evaluated for infertility.
- Unilateral: the normal testis carries fertility, so semen overstates infertility, and paternity after removing the affected testis equals controls. Bilateral: semen reflects true infertility.
- Ad spermatogonia at biopsy predict adult counts (present on both sides: normal count in 84%; absent: all subnormal). Ad per tubule > 0.2 beats S/T. In bilateral cases INHB/FSH and AMH/FSH ratios aid counselling.
- Even after first-year surgery, 20–25% remain at infertility risk.
- Surgery **before 18 months** reduced assisted reproduction; each 6-month delay added 5%.
- Orchidopexy before 19 years improves sperm retrieval in azoospermia; retrieval beats idiopathic non-obstructive azoospermia.
- Uncomplicated retractile testis: insufficient evidence of infertility.

5 · Malignancy

Why this matters: orchidopexy lowers cancer risk only partly.

- TGCT risk is **2–5×** (pooled odds ratio (OR) ≈ 4). About 1% of cryptorchid men get TGCT; 5–10% of TGCT patients had a UDT. Contralateral OR ≈ 1.5.
- In the **Swedish population study** the OR was 3.2: about 2.8 if diagnosed before puberty vs about 5.0 after. † Pettersson 2007: relative risk (RR) 2.23 if operated before 13 years vs 5.40 at 13 or older. A meta-analysis gave RR **5.8** for orchidopexy after 10–11 years.
- Untreated adults have OR ≈ 14; risk rises with each 6-month delay beyond 18 months. Early surgery reduces only the positional component.
- A persistently cryptorchid testis develops **seminoma** (74%); after orchidopexy tumours are mostly non-seminoma (63%).
- Germ cell neoplasia in situ (GCNIS) comes from gonocytes that fail to differentiate. It occurs in 2–3% of formerly cryptorchid men and ≤ 0.4% of non-syndromic children, so biopsy DSD and chromosomal cases. Contralateral GCNIS occurs in 4–8% of men with TGCT (predictors: atrophy, cryptorchidism, age < 40).
- Biopsy at orchidopexy does not raise TGCT risk (earlier cohort confounded; a Scandinavian universal-biopsy cohort found no excess).
- Postpubertal UDT with a normal contralateral testis: **orchidectomy** (up to age 50) with a prosthesis at the same sitting.
- **Microlithiasis** occurs in 3–6% of boys. It is not an independent TGCT risk, so routine ultrasound (US) surveillance is not needed.

6 · Diagnosis

Why this matters: examination decides; imaging adds nothing; bilateral non-palpable testes may mean CAH.

Examination — the gold standard

ALGORITHM

1. Warm room and hands, distraction. Examine supine, then cross-legged, squatting or standing; thigh abduction inhibits the cremaster.
2. With lubricant on the fingers, sweep one hand from the anterior superior iliac spine (ASIS) along the canal; palpate the lowest position with the other.
3. Compress the canal and invaginate the scrotum to feel the SIP.
4. Apply sustained traction to fatigue the cremaster. If it **stays, it is retractile; if it springs back, it is undescended**.

- Document palpability, position, size, mobility, lie, hernia/hydrocele, penile size and meatus. Scrotal asymmetry supports unilateral UDT.
- About **80% palpable**; 60–70% are unilateral. At surgery a non-palpable testis is abdominal/peeping in 25–50%, vanished in 15–40% and a missed extra-abdominal testis in 10–30%.
- Interobserver agreement is poor; examination under anaesthesia (EUA) is the final arbiter.
- P3 P1 Maldescended (undescended) testes and treatment algorithm; Vanishing testes — *AUA algorithm: bilateral non-palpable, then DSD; re-examine at 6 months corrected age; EUA, then yearly review / orchidopexy / exploration / laparoscopy; contralateral hypertrophy > 2 mL (specificity 100%) suggests monorchia, explore anyway*

Not in the paper answer:

- Paternity after unilateral UDT equals controls (about 90%; the paper says four-fifths).
- Seminoma if left cryptorchid (74%) vs non-seminoma after orchidopexy (63%); orchidectomy advised up to age 50.

Imaging — not recommended

American Urological Association (AUA) 2014: no US or magnetic resonance imaging (MRI). US sensitivity is about 45% and specificity 78%; MRI 65%/100% but needs sedation. Neither avoids nor excludes surgery; the only effect is about 3 months' delay. MRI may help with an ectopic abdominal testis not found at laparoscopy.

Bilateral non-palpable or UDT with hypospadias: DSD work-up

- A newborn with bilateral non-palpable testes needs an **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)**, because of salt-wasting. Pelvic US looks for Müllerian structures; defer circumcision.
- Hypospadias coexists in 12–24%; proximal hypospadias with UDT carries a chromosomal abnormality in 32–47% *(see Ch 56)*.
- Micropenis with bilateral UDT suggests hypogonadotrophic hypogonadism or anorchia (micropenis in nearly half). Measure LH, FSH and testosterone during minipuberty *(see Ch 55)*.

Marker	Anorchia	Bilateral abdominal UDT
FSH, LH	Raised	Normal
AMH, INHB (best prepubertal Sertoli markers)	Undetectable/low	Present

Marker	Anorchia	Bilateral abdominal UDT
Basal testosterone	Low in first year only	Normal
Testosterone after hCG	No rise	Rises

hCG stimulation is poorly standardised and no longer the test of choice; gonadotrophins may be normal in mid-childhood anorchia. **Laparoscopy remains the confirmatory test**. In a genetic male with bilateral non-palpable testes, ≥ 95% are abdominal.

7 · Management principles and timing

Why this matters: early surgery gives the best outcomes; hormones cannot replace it.

- **Timing**: observe to **6 months** (corrected age), then refer; operate at **6–18 months**, preferably by 12. A preterm boy is observed to 6 months beyond the due date (1 year if marginal).
- Abdominal orchidopexy is easier soon after minipuberty. Most children worldwide still have surgery after 2 years.
- Retractile: no surgery, yearly examination. Acquired UDT: operate, as no paternity or cancer data support observation, especially if bilateral.
- Pubertal or postpubertal with a normal contralateral testis: orchidectomy (laparoscopic if abdominal), with or without a prosthesis.

Hormonal therapy — not recommended (AUA)

- Intranasal luteinising hormone-releasing hormone (LHRH) gives descent in about 20%; hCG was never placebo-tested. Neither separates retractile from undescended testes.
- hCG may increase germ cell apoptosis and reduce adult volume (infertility RR 4.7 after hCG in one series).
- Adjuvant LHRH (buserelin 10 µg intranasally on alternate days for 6 months; Hadziselimovic) for bilateral UDT with absent Ad spermatogonia ("defective minipuberty") has pilot data only and needs a biopsy.

> **KEY POINT**
> Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable — descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

8 · Surgery for the palpable testis

Why this matters: the goal is a tension-free scrotal testis with intact vessels and a ligated processus.

Inguinal orchidopexy — key steps

ALGORITHM

1. EUA, skin-crease incision, check the SIP.
2. Open the external oblique, sparing the ilioinguinal nerve.
3. Free the testis; divide the gubernaculum distal to the sac (protect a long-looping vas).
4. **Separate the patent processus from vas and vessels without skeletonising; high-ligate it at the internal ring** (main length gain).
5. If short, divide lateral spermatic bands and dissect retroperitoneally through the transversalis fascia; then the **Prentiss manoeuvre** (medial to the inferior epigastric vessels).
6. Transverse scrotal incision and **subdartos pouch**; pass the testis untwisted.
7. Excise appendages, inspect the epididymis, close dartos around the cord (transalbugineal sutures unnecessary — 98.7% success without). Local, ilioinguinal or caudal block.

- If the testis cannot reach the scrotum, stage the procedure. Orchidectomy is only for an atrophic or abnormal testis, a postpubertal boy, or an insufficient vas.
- Biopsy is not routine; it is indicated in DSD/dysgenesis and considered after 12–13 years and in bilateral UDT.

Scrotal (Bianchi) orchidopexy

Through a high scrotal-border incision, deliver the testis, dissect sac and cremaster high, ligate the processus and fix in a subdartos pouch. Inguinal conversion is 3.5–4.4% and atrophy 0.3–0.6%. With less pain and similar efficacy, it is increasingly first choice for palpable, especially low, testes.

9 · The non-palpable testis

Why this matters: laparoscopic findings at the ring decide the next step.

ALGORITHM

1. EUA: if now palpable, do an inguinal or scrotal orchidopexy.
2. If **still non-palpable, do diagnostic laparoscopy** (ring patency, vessels, vas, testis).
3. If **blind-ending vessels** (with or without vas) lie above a closed ring, the testis has vanished: stop (the only unequivocal endpoint).
4. If **vessels and vas enter the ring**, explore the groin/scrotum and excise the nubbin (viable tubules 5–15%).
5. If the testis is **intra-abdominal**, do a primary laparoscopic or Fowler–Stephens (one- or two-stage) orchidopexy; orchidectomy if postpubertal or dysgenetic.
6. If it is peeping at the ring, do a laparoscopic or inguinal orchidopexy.
7. If neither vas nor vessels are seen, search the perivesical area and retroperitoneum to the kidney (a perirenal testis may accompany renal dysplasia/agenesis).

- Pitfalls: an open ring usually (not always) means a viable distal testis. Apparently blind-ending vessels may coexist with a distal or ectopic testis; if equivocal, keep exploring.
- Scrotal-first exploration (nubbin with contralateral hypertrophy) is futile if the vanished testis is intra-abdominal, and the "nubbin" may be a dissociated epididymis.
- Fixing the contralateral solitary testis is debated: prenatal torsion does not imply contralateral risk, and the AUA does not require it.

Laparoscopic orchidopexy

ALGORITHM

1. Empty bladder and stomach. Open (Hasson) umbilical 5-mm port; carbon dioxide (CO_2) 8–12 mmHg. Inspect both rings; add 2–3-mm working ports.
2. Divide the gubernaculum distal to the ring and use it as a handle.
3. Incise peritoneum lateral to the vessels and distal to the vas; mobilise the vessels, **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible).
4. Make a new hiatus medial to the inferior epigastric vessels and lateral to the medial umbilical ligament.
5. Bring the testis through a trans-scrotal port to a subdartos pouch; avoid tension.

- Reaching the opposite ring is an unreliable test of length. Ring closure is unnecessary. Contralateral patent processus ≈ 9%; repair optional.
- Laparoscopy also suits bilateral cases, abdominal wall defects, polyorchidism, splenogonadal fusion and transverse ectopia. The robot is feasible, with no clear advantage.

Fowler–Stephens (FS) and alternatives

- P1 Methods of orchidopexy — *FS on vasal/gubernacular collaterals for testes not within 2–4 cm of ring; success table: primary laparoscopic > 90–97%, Jones/Bagley ≈ 95%, FS 60–97% (one-stage 63%, two-stage 68%), Shehata 84–85% (atrophy 10%), autotransplant 96%; Dessanti PTFE traction*

Not in the paper answer:

- Clip level, gubernacular sparing and the germ cell cost of dividing vessels (below).
- Laparoscopic complications: bladder/vessel injury, hypercapnia, bowel obstruction.
- Two-stage FS: clip the vessels high (1.5–3 cm above the testis), wait about **6 months**, then mobilise on a wide peritoneal pedicle. Two-stage is slightly favoured (better position, similar atrophy).
- Keep peritoneum over the vasal vessels and **spare the gubernaculum** (collaterals in 60%); this lowers atrophy.
- Avoid dividing vessels when possible, because it reduces germ cells, especially in young boys. A retrovesical testis with a short vas may need orchidectomy.
- Shehata: the testis is fixed near the contralateral ASIS and brought down after 12 weeks, keeping the artery. Autotransplantation suits a solitary abdominal testis but needs microsurgical skill.
- Bilateral high testes: operate on the two sides about 6 months apart.

KEY POINT

Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it — vessel division is the last resort.

10 · Complications and follow-up

Why this matters: atrophy and re-ascent are the main failures, and technique prevents most.

Complication	Cause / prevention	Management
Atrophy (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal)	Vessel injury, skeletonisation, FS, tension	Offer a **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos
Retraction / non-scrotal (≈ 2–10%)	Inadequate retroperitoneal mobilisation, unligated processus	Redo: a prescrotal testis by the scrotal approach; otherwise a lateral approach through unscarred tissue (Redman) or a fascial patch on the cord (Cartwright). Results equal primary surgery.
Vas injury	Long-looping vas cut with the gubernaculum	Divide the gubernaculum distal to the sac; avoid skeletonising
Torsion after orchidopexy	Rare	Prevented by subdartos fixation
Laparoscopic	Bladder/vessel injury, hypercapnia, bowel obstruction	Hasson entry, bladder drainage

Follow-up: examine at ≥ **6 months** for position and size (some add US after FS). Counsel on fertility and cancer, and teach self-examination from puberty.

61 Urologic Considerations in Pediatric Renal Transplantation

- P1 Urological evaluation and management of pediatric kidney transplant patient. — *causes, five goals, pre-transplant screen (VCUG/urodynamics indications), bladder optimisation and cycling, augment segments, transplant technique, complication rates and management*

Not in the paper answer:

- Graft reflux costs more than native reflux (solitary immunosuppressed kidney).

1 · Causes of ESKD in children

Why this matters: the bladder that destroyed the native kidneys can destroy the graft.

- Commonest cause of end-stage kidney disease (ESKD): congenital anomalies of the kidney and urinary tract (**CAKUT**: hypoplasia/dysplasia), then obstructive uropathy, mainly posterior urethral valves (**PUV**) in younger boys.
- Older children: focal segmental glomerulosclerosis (FSGS), neuropathic bladder. Commonest age at transplant: **13–17 years**.
- The urologist joins the team **before ESKD**, not reactively.

> **KEY POINT**
> **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

2 · Pre-transplant evaluation

Why this matters: most children need only a simple screen.

- All: history, renal ultrasound (US), and uroflow with post-void residual (PVR) if toilet-trained.
- Voiding cystourethrogram (VCUG) and urodynamics are selective (neuropathic bladder, PUV, vesicoureteral reflux (VUR), prior surgery, febrile urinary tract infection (UTI), worsening hydronephrosis); urodynamics are repeated.
- Anuric child without urological history: nothing more.

> **ALGORITHM**
> 1. Positive history: US and VCUG. If dynamics are abnormal, urodynamics.
> 2. Treat with drugs and clean intermittent catheterisation (CIC), with or without bladder cycling, then reassess.
> 3. If still unsafe, augment; then clear for transplant.

3 · Bladder management

Why this matters: the bladder must be safe before the graft goes in.

- Commonest pattern: **low capacity, overactivity, poor compliance**; high pressure harms the kidney even without reflux. A valve bladder evolves to **myogenic failure**.
- **Target**: expected capacity at detrusor pressure **< 40 cm H_2O**, with social continence.
- Drugs: antimuscarinics, mirabegron, botulinum toxin; α-blockers aid emptying. Better storage may need CIC, and CIC adherence predicts transplant-regimen adherence.

- **Defunctionalised bladder**: bladder cycling (rising saline volumes with drugs, then urodynamics). **Never use the allograft to test a bladder.**
- **Augment** if unable to reach > **75%** of expected capacity at < 40 cm H_2O with CIC no more often than every **3–4 h** despite maximal therapy. It does not clearly raise transplant complications. PUV: **postpone**; neuropathic: **earlier**.
- Segments: dilated ureter first (no mucus, stones or metabolic issues). Ileum/sigmoid are most used; malignancy is worse with immunosuppression, so surveillance is lifelong. Stomach is rare (haematuria–dysuria in anuric children).
- Site a **Mitrofanoff medial to the planned transplant incision**. Preserve the native ureter for a stoma, augment or graft drainage.
- Reconstruct **before** transplant, allowing **6–12 weeks** to heal; exception: transplant before 2 years (before continence is expected).
- On peritoneal dialysis (PD), intraperitoneal surgery forces temporary haemodialysis. Avoid concurrent procedures in infants. Long-term stoma or transplant ureterostomy is only for severe neurodevelopmental impairment or CIC intolerance.
- Circumcision in boys at risk (valves) reduces post-transplant UTI.

4 · Native nephrectomy

Why this matters: native kidneys stay unless there is a named indication.

- P2 Pre transplant nephrectomy — *default keep; indications (hypertension, nephrotic, polyuria, infection/VUR, stones, malignancy, size); retroperitoneal on PD, embolisation if high risk; leave normal ureter, remove dilated refluxing ureter*

Not in the paper answer:

- Combine it with the transplant only to make room (small child, huge kidneys), ipsilaterally through the transplant incision.

5 · Transplant into the abnormal bladder

Why this matters: the ureteric join must neither reflux nor obstruct.

- Gibson retroperitoneal incision; a child < **15 kg** may need the aorta/vena cava and a transperitoneal approach. Know where the augment pedicle lies.
- Preferred: **extravesical non-refluxing ureteroneocystostomy** (partly fill bladder, 3–4 cm detrusorotomy, spatulate 4–5 mm, interrupted mucosa-to-mucosa, close flaps loosely). Alternatives: Barry tunnel; **ureteroureterostomy** to the native ureter (small bladder, short donor ureter, hostile pelvis), always the salvage.
- Stent if difficult: double-J out at **4–6 weeks**, and early removal at 5–7 days may cut infection. Use the largest urethral catheter tolerated.

6 · Urological complications

Why this matters: the graft tolerates obstruction poorly.

Complication	Rate	Timing / management
Ureteric stenosis	**6%**	Most within 100 days; distal, ischaemic. Focal: balloon and stent for 4–6 weeks; otherwise open repair (native ureter, bladder flap); do not delay.
Urine leak	**3%**	Early. Drain-fluid creatinine; check catheter, then US, then MAG3 (mercaptoacetyltriglycine) scan or cystogram. Small: observe; large: drain or explore. Short necrosis: reimplant; long: native ureter, psoas hitch or flap.

Complication	Rate	Timing / management
Significant VUR	2%	See below
Stones	2%	Treat if > 1 cm and growing, obstructing or infected; one-stage modality

MAG3 localises a leak only if graft function is adequate. Pre-existing obstructive or reflux uropathy raises all complications.

Rising creatinine with hydronephrosis

> **ALGORITHM**
> 1. Indwelling catheter for 1–2 weeks: if creatinine falls, the bladder is the cause.
> 2. If not, stent and keep the catheter; biopsy if rejection is suspected.
> 3. If retrograde stenting fails, nephrostomy and antegrade study.

- The graft is unusually sensitive to obstruction; hydronephrosis grade does not track impairment. Obstruction and rejection coexist, so stent diagnostically at a low threshold.
- Extrinsic: lymphocele; an **allograft mass is post-transplant lymphoproliferative disorder (PTLD) until proven otherwise**.
- Early UTI comes from catheter or stent; late UTI reflects bladder function. Colonisation is universal with CIC; distinguish it from infection.
- Graft VUR: febrile UTI needs a cystogram; recurrent febrile UTI with an optimised bladder needs an **open/robotic reimplant** (intravesical, transtrigonal or Politano–Leadbetter; stent). Observe asymptomatic reflux. Endoscopic injection succeeds in about one-third and risks obstruction.

7 · Outcomes

Why this matters: repeated infections erode graft function.

- Urological complications rarely lose the graft; thrombosis is a leading cause of early loss.
- **≥ 2 UTIs** lower graft glomerular filtration rate (GFR) at 1 and 4 years, the gap widening.
- A PUV history lowers obstruction-free survival; its link with graft dysfunction is inconsistent.

63 Paediatric Urologic Oncology: Renal and Adrenal

1 · Wilms tumour (WT): epidemiology and genetics

Why this matters: genetics decide who is screened and who needs kidney-sparing surgery.

- P2 Wilms tumour: classification, staging and management — *genes (WT1 11p13, IGF2 11p15), syndrome risks, screening, anaplasia, SIOP risk groups, COG staging, COG vs SIOP, nodes, adjuvant by stage*

Not in the paper answer:

- Spill rate (~10%), complication rate (~11%), relapse salvage groups and late-effect figures (sections 5–6).
- Caval thrombus: en bloc removal below the hepatic veins; piecemeal removal means stage III.
- WT is **6–7%** of childhood cancers and **95%** of kidney cancers under 15 y; median age **3.5 y**, younger if bilateral/syndromic.
- Most arise from several somatic hits (Knudson two-hit explains few); **10–15%** are germline/early epigenetic; 1–2% familial.
- **Refer for genetics**: bilateral/multifocal, age < 2 y, rests, overgrowth/syndromic features, family history.
- WTX (Xq11) 20% and CTNNB1 15% are somatic only; TP53 4% (mostly anaplastic).
- WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) is a WT1 deletion with renal-failure risk into puberty. **Denys–Drash** syndrome (DDS) is a zinc-finger point mutation, **dominant-negative** and so worse, with early end-stage renal disease (ESRD).
- **Type I** (WT1 ± CTNNB1): stromal, intralobar rests, early age, WAGR/DDS. **Type II** (11p15): blastemal/epithelial, perilobar rests, later, Beckwith–Wiedemann syndrome (BWS).
- In BWS 21% are bilateral, and hepatoblastoma and adrenocortical tumours occur; nephromegaly (≥ 95th centile) carries the highest risk.
- Combined 1p and 16q loss of heterozygosity (LOH) (~5%) earns the intensified **regimen M**; 1q gain (25–30%) is the next stratifier.

Syndrome risk	Examples
High (> 20%)	WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy
Moderate (5–10%)	BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3)
Low	Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom

KEY POINT

Trap: BWS is the **commonest** predisposing syndrome but only moderate risk; WAGR/DDS are rarer and highest.

ALGORITHM

1. If WT risk is > 1%, do a renal ultrasound (US) every **3–4 months** until **7 y**.
2. If a lesion looks suspicious, do computed tomography (CT) or magnetic resonance imaging (MRI).

- BWS, hemihypertrophy, trisomy 18, Simpson–Golabi–Behmel: add α-fetoprotein (AFP), 3-monthly to 4 y (hepatoblastoma).
- After nephrectomy, scan the other kidney 3–4-monthly to 5 y (infants < 12 months with rests).

- Screening lowers stage and allows nephron-sparing surgery (NSS); survival benefit is unproven. Benign cysts are common in BWS; avoid needless nephrectomy.

2 · Pathology

Why this matters: anaplasia marks chemoresistant tumours.

- Favourable histology is **triphasic**. The tumour is soft and friable, so it spills easily; 12% are multicentric in one kidney.
- **Anaplasia**: nuclei ≥ **3×** neighbours, hyperchromasia, abnormal mitoses. Seen in ~2% under 2 y and ~13% over 5 y; acquired (absent from rests).
- Nephrogenic rests are found in > 1/3 of WT kidneys and **> 90%** of bilateral cases; most regress.
- Diffuse hyperplastic perilobar nephroblastomatosis forms a thick rind, is almost always bilateral, and half develop WT.
- Perilobar rests: periphery, late, BWS. Intralobar: anywhere or sinus, early, WAGR/DDS.

3 · Clinical features and work-up

Why this matters: simple clues predict caval thrombus and bleeding.

- > 85% are symptomatic. Gross haematuria needs cystoscopy and retrograde study at surgery (collecting-system extension).
- A varicocele persisting supine, or hepatomegaly, suggests inferior vena cava (IVC) thrombus.
- **Acquired von Willebrand disease** occurs in up to 8% (clotting times may be normal); give desmopressin.
- Do chest CT before MRI anaesthesia (atelectasis). No contralateral exploration if imaging is normal.
- Ascites beyond the cul-de-sac best predicts rupture; emergency surgery only for bleeding. Imaging cannot judge nodes.
- Imaging cannot separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK); CCSK and rhabdoid need bone scan and brain imaging.

4 · Staging and COG vs SIOP

Why this matters: COG operates first; SIOP gives chemotherapy first.

- Children's Oncology Group (COG) stages I–V are surgical–pathological. Stage III means **positive nodes, any spill**, implants, a positive margin or piecemeal removal.
- The International Society of Paediatric Oncology (SIOP) gives pre-operative vincristine (VCR) and dactinomycin (AMD) without biopsy, with US and chest radiograph follow-up (COG: CT).

> **KEY POINT**
> COG stage is surgical-pathological, so the surgeon — not the scan — decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

5 · Surgical principles

Why this matters: the operation sets the stage and hence treatment.

- Transperitoneal radical nephrectomy; inspect liver, peritoneum and nodes; intra-operative US if vein or IVC is equivocal.

- Sample nodes (COG **≥5 nodes**; skip metastases occur interaortocaval). Omission is the commonest protocol violation (10–20%) and under-stages.
- Spill (~10%; right side, large tumours) causes local relapse. Local recurrence risks: spill, unfavourable histology, incomplete removal, no node sampling.
- Complications ~11% (haemorrhage, small-bowel obstruction); more with > 10 cm, caval extension, thoracoabdominal incision, wrong diagnosis.
- Chemotherapy: cyclophosphamide and etoposide are added in regimen M and for CCSK; carboplatin for anaplastic or relapsed disease. † COG favourable histology: stage I–II get VCR and AMD; stage III–IV add doxorubicin (DOX) with flank/abdominal (and lung) radiotherapy (RT).

Bilateral WT (stage V) and NSS

- P3 Bilateral Wilms tumour: treatment strategies — *5–7%; no upfront nephrectomy/biopsy; 3 drugs 6 weeks, then NSS by week 12 (NWTS-4 anaplasia); open bilateral biopsy (discordant 4/5); enucleation + flank RT; transplant after 1–2 y; AREN0534 EFS 82%, OS 95%*

Not in the paper answer:

- Remove residual renal tissue before transplant; in NSS use frozen section and a normal-kidney margin.

> **ALGORITHM**
>
> 1. Synchronous bilateral WT: **no upfront nephrectomy, no biopsy**.
> 2. Give VCR, AMD and DOX for 6 weeks, then re-image.
> 3. If NSS is feasible, operate.
> 4. If not, give more chemotherapy, but do **surgery by week 12**.
> 5. If the tumour does not respond, do a **bilateral open biopsy**.

- The commonest cause of renal failure is bilateral nephrectomy for persistent tumour.
- Unilateral NSS suits syndromic children or a solitary kidney; UMBRELLA criteria: **<300 mL**, adequate remnant, negative nodes.

IVC thrombus

- ~6–10%; over half asymptomatic. Below the hepatic veins: upfront nephrectomy with **en bloc** thrombectomy (piecemeal means stage III).
- **At or above the hepatic veins: give pre-operative chemotherapy** to avoid bypass; chemotherapy beyond 6–12 weeks adds nothing. Viable cells in the thrombus do not affect survival.

6 · Relapse and late effects

Why this matters: salvage depends on prior treatment.

Relapse group	Definition	Salvage
Standard	Non-anaplastic, previously VCR/AMD only	70–80%
High	Non-anaplastic, after ≥ 3 drugs	40–50%
Very high	Anaplastic/blastemal	~10%

- Late effects track **RT and anthracycline**: heart failure 4.4% after DOX (rare below **150 mg/m²**), second cancers 1% at 10 y, gonadal failure.
- Renal failure after unilateral nephrectomy is only **0.6%** at 20 y (higher in WT1 syndromes).

7 · Other renal tumours

Why this matters: non-Wilms tumours need different treatment.

Tumour	Key facts	Management
Congenital mesoblastic nephroma (CMN)	Commonest **neonatal** renal tumour (mean 3½ months); ETV6–NTRK3 fusion in the cellular type	Radical nephrectomy alone; adjuvant only if incompletely resected
Clear cell sarcoma (CCSK)	3%; BCOR; **bone** and brain metastases; late relapse (to 10 y); never bilateral	DOX-containing chemotherapy; RT even at stage I
Rhabdoid tumour (RTK)	2%; **most lethal**; median ~16 months; SMARCB1/INI1 loss (22q); associated brain tumours	No standard; tazemetostat on trial
Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN)	Look identical on imaging; CPDN has blastema in the septa; DICER1	Excision cures; frozen section if partial nephrectomy
RCC	RCC outnumbers WT above 13–14 y. **Translocation RCC (TFE3/TFEB, Xp11.2)** is commonest. Medullary carcinoma occurs with sickle trait and is lethal.	Complete resection; nodes less ominous than in adults; resistant to chemotherapy and RT
Angiomyolipoma	Tuberous sclerosis (50–80%); bleeding risk if > 3 cm and growing	Mammalian target of rapamycin (mTOR) inhibitor first; embolise for bleeding

KEY POINT

Age points to diagnosis: a neonate has CMN, a toddler WT or rhabdoid, an adolescent RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

8 · Neuroblastoma

Why this matters: biology, not radical surgery, decides outcome.

- Commonest **extracranial solid tumour** and commonest infant malignancy; neural crest origin; median 19 months; 50% adrenal; ~70% metastatic at diagnosis. Familial cases are < 1% (PHOX2B, ALK).
- The spectrum runs through ganglioneuroblastoma to **ganglioneuroma** (benign; operate only for symptoms). Shimada: stroma-poor vs rich, graded against age.
- Presents as a hard fixed mass, bone pain, periorbital ecchymosis or cord compression; pelvic tumours cause retention.
- Paraneoplastic: catecholamines; vasoactive intestinal peptide (VIP) causing watery diarrhoea; **opsoclonus–myoclonus–ataxia** (half have neuroblastoma).
- Urinary vanillylmandelic acid and homovanillic acid (**VMA/HVA**) are raised in **90–95%**. Marrow: two aspirates and two biopsies. MRI is best for intraspinal extension.
- **Calcification and vessel encasement** favour neuroblastoma over WT.
- ^{123}I-meta-iodobenzylguanidine (MIBG) is required for staging (replaces bone scan); non-avid tumours need positron emission tomography (PET); ^{131}I-MIBG is therapeutic.
- Screening found more favourable infant tumours without fewer advanced cases or deaths — **overdiagnosis**.

International Neuroblastoma Risk Group (INRG) stage	Definition
L1	Localised, **no** image-defined risk factor (IDRF)
L2	Localised, ≥ 1 IDRF (vessel/nerve encasement)
M	Distant metastases
MS	Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months)

- INRG is imaging-based, before treatment; the International Neuroblastoma Staging System (INSS, 1–4S) depends on resection and nodes, so it is surgeon-dependent.
- Risk groups use stage, age, histology, MYCN, 11q and ploidy. **MYCN amplification** (> 10 copies, 2p24; ~20%) means high risk at any stage (except maturing histology). M at ≥ 18 months is always high risk, as is MS with 11q aberration.
- Adverse features: 1p36 deletion, 11q LOH, 17q gain (strongest) and **diploidy**. Favourable: **hyperdiploidy**, TrkA expression, age < 18 months and a thoracic site.

> **KEY POINT**
> Trap: diploid is bad, hyperdiploid good. Stage 4S/MS: OS 80–88%, frequent regression; deaths in infants < 2 months from massive hepatomegaly (respiratory compromise) or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemotherapy only for symptoms, progression or adverse biology. Never remove the kidney or vital structures; RT to stage 2 residual does not help.
- **4S**: resecting the primary is not mandatory; asymptomatic favourable tumours get supportive care with or without limited chemotherapy.
- **Intermediate**: chemotherapy, then resection (fewer nephrectomies).
- **High risk**: induction (cyclophosphamide, vincristine, doxorubicin, alternating with cisplatin–etoposide), then delayed surgery after 3–4 courses, myeloablative consolidation with autologous stem-cell rescue (tandem is better), RT 15–30 Gy, and **13-cis-retinoic acid with anti-GD2 antibody** (plus GM-CSF and interleukin-2).
- Stage 4: radical resection gives no survival benefit but better local control; delayed surgery means fewer ruptures and nephrectomies. Complications: diarrhoea (autonomic nerves cut at coeliac/superior mesenteric origins), chylous ascites.
- **Cord compression**: chemotherapy first; laminectomy only for progressive deficit (it causes scoliosis); avoid RT.

> **KEY POINT**
> Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection — vital structures are never traded for margin.

9 · Adrenocortical tumours

Why this matters: they are often hereditary and cured only by intact removal.

- P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *bimodal (children < 10 y, 90% functional); Li–Fraumeni/BWS; open en bloc, intact capsule; mitotane + EDP*

Not in the paper answer:

- In children virilisation is commonest (cortisol in adults); TP53 testing for all; weight > 200 g is adverse.
- † Offer **TP53** testing and counselling to every child. Most are functional and under 5 y: androgens cause **virilisation** (commonest), cortisol Cushing syndrome (often mixed), oestrogen feminisation; aldosterone excess is rare.
- † Work-up: dehydroepiandrosterone sulphate (DHEA-S), testosterone, androstenedione, 17-hydroxyprogesterone, cortisol tests, a **urinary steroid profile** and metanephrines; CT/MRI and chest CT.
- † Adverse: weight > 200 g, incomplete resection/spill, metastases, Cushing features. Adenoma vs carcinoma is hard to separate in children.
- † Open en bloc, **avoid capsular rupture**, no laparoscopy if carcinoma suspected; node dissection if large; **perioperative steroids** for cortisol-secreting tumours.

> **KEY POINT**
> Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

10 · Landmark trials

Why this matters: these trials set current practice.

Trial	Design	Result	Take-home
AREN0532/0533	1p/16q LOH treated with regimen M	Stage III–IV event-free survival (EFS) ~90% vs 61%	Biology-driven intensification
AREN0533 (lung)	Lung nodule response at week 6	RT omitted, EFS and overall survival (OS) preserved	Response-adapted RT omission
AREN0534	Bilateral WT: 3-drug, surgery by week 12	EFS 82%, OS 95% (vs 56%/81% in NWTS-5)	Fixed clock, maximise NSS
COG P9641	Low-risk neuroblastoma, surgery alone	EFS ~89%, OS ~97%	Chemo only if symptomatic
COG anti-GD2	Anti-GD2 + cytokines + retinoid vs retinoid	Antibody superior	Immunotherapy standard in high risk
German NB97	Extent of resection, stage 4 > 18 months	No EFS/OS benefit	Radical resection not essential

64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

1 · Rhabdomyosarcoma (RMS): biology

Why this matters: fusion status, not the microscope, now sets risk and treatment.

- RMS is the **commonest childhood sarcoma** (~3% of paediatric cancers); 20–30% are genitourinary (GU). It peaks in the first 2 y and in adolescence.
- GU sites: bladder/prostate (BP), paratesticular (PT), vulvovaginal (VV). GU-RMS 5-y survival is **>80%**; 16% are metastatic at diagnosis.
- ≥ 5% are syndromic, usually embryonal: **Li–Fraumeni**, Beckwith–Wiedemann (BWS), **DICER1**, neurofibromatosis 1 (NF1, pelvic) and Costello (bladder). Diagnosis < 3 y should prompt a germline search.
- Surveillance: Li–Fraumeni, ultrasound (US) 3–4-monthly and yearly whole-body/brain magnetic resonance imaging (MRI); DICER1, chest computed tomography (CT) and US; Costello, US 3-monthly to 8–10 y.

	Embryonal (ERMS)	Alveolar (ARMS)
Site	GU (commonest); the **botryoid** variant protrudes from bladder or vagina	Trunk, extremities, perineum
Age	Younger (~70% < 10 y)	Older, adolescents
Fusion	Usually negative	~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)]
Prognosis	Better	Fusion-positive worse; fusion-negative ARMS behaves like ERMS

KEY POINT

Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**. MYOD1 mutation and circulating tumour DNA are emerging adverse markers.

2 · Presentation and work-up

Why this matters: staging scans follow the nodal drainage of each site.

- BP: haematuria, stranguria, retention, or a botryoid mass at the meatus. PT: a painless mass separate from the testis. VV: bloody discharge or a prolapsing mass.

ALGORITHM

1. Start with US.
2. CT or MRI of the primary and its nodal basin: BP, pelvic and retroperitoneal nodes up to the renal vessels; PT, retroperitoneal; VV, inguinal, pelvic and retroperitoneal.
3. **Chest CT** (lung is the commonest metastatic site).
4. Bone scan and bilateral marrow sampling; omit these in fusion-negative ERMS without nodal or lung disease.

3 · Stage, group and risk

Why this matters: three layers set treatment intensity.

The **stage** is pretreatment tumour–node–metastasis plus site. The **clinical group** records what the surgeon left. The **risk group** adds fusion status and age.

Children's Oncology Group (COG) stage	Site	Size / nodes
1	Favourable: **paratesticular, female GU**, orbit, head–neck	Any, M0
2	Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk	≤ 5 cm, N0/NX
3	Unfavourable	≤ 5 cm N1, or > 5 cm any N
4	Any	M1

N1 means a node ≥ 1 cm in short axis. BP-RMS is never below stage 2.

Clinical group	Definition
I	Localised, completely resected, nodes negative
IIa / IIb / IIc	IIa: microscopic residual. IIb: involved nodes resected, most distal node negative. IIc: microscopic residual or most distal node positive.
III	Gross residual or **biopsy only**
IV	Distant metastases

Risk	Criteria
Low	Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II
Intermediate	Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y
High	Stage 4 fusion-positive, or ≥ 10 y

KEY POINT

An organ-preserving **biopsy** places BP-RMS in group III, which means **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y is independently adverse. Trap: most distal node positive = IIc, not IIb.

4 · Treatment principles

Why this matters: chemotherapy does most of the work, so surgery aims to save the organ.

- **Chemotherapy is universal**; local control is by radiotherapy (RT) and/or surgery, and the group sets RT dose. Complete resection (except PT) only if the organ is preserved.
- Primary exenteration fell from almost all cases (Intergroup Rhabdomyosarcoma Study, IRS-I) to **13%** (IRS-IV), with overall survival (OS) unchanged (78–83%).
- **Pretreatment re-excision** (before chemotherapy; resectable without organ loss; positive or uncertain margins) improves GU survival and can change the group.
- **Delayed primary excision** (after chemotherapy starts) has no proven survival gain and cannot change the group; do it only if a negative-margin (R0) resection is likely. It may lower RT dose.

Bladder/prostate RMS

- Biopsy: **cystoscopic** preferred (resectoscope, minimal cautery), then percutaneous; open transvesical last (spill).
- Obstruction: **urethral catheter** first, then JJ stent, then nephrostomy. Avoid a suprapubic tube (seeding); vesicostomy or ureterostomy delays chemotherapy.
- An **end-of-therapy residual mass** is usually stroma or mature rhabdomyoblasts, so observe it if stable; resection does not improve survival.
- Viable tumour after chemoradiation, M0: partial cystectomy or cystoprostatectomy/exenteration can cure.
- Only ~40% of preserved bladders are "normal": **urodynamics at 6 months**, renal US yearly for ≥ 10 y.

Paratesticular RMS

- **Radical inguinal orchidectomy**. After a scrotal approach, excise the remaining cord inguinally; hemiscrotectomy or scrotal RT only for a positive cord margin or skin invasion.
- A tumour abutting but not invading the tunica vaginalis is **not** a positive margin.
- Do an **ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or nodes are radiologically N1. Remove ≥ 7–13 nodes up to the renal vessels.

> **KEY POINT**
> Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

Female GU, perineal, infants

- Vaginoscopy and biopsy. Chemotherapy with or without brachytherapy controls most; no node dissection; hysterectomy/exenteration only if unresponsive. Cervical: loop biopsy, repeated after induction.
- Vulvovaginal (two-thirds of female GU): MRI and vaginoscopy after 3 and/or 6 cycles; positive residual gets brachytherapy.
- Perineal/perianal: mostly alveolar fusion-positive; 5-y OS ~50%.
- Infants: RT is often withheld, so local failure is commoner; RT deviation is allowed in fusion-negative children < 24 months.

Late effects: RT causes about tenfold more sequelae; bladder dysfunction > 60%; sexual dysfunction, infertility, ovarian failure, vaginal stenosis; second cancers ~6×.

5 · Other bladder tumours

Why this matters: most non-RMS bladder lesions in children are low grade or benign.

- **Urothelial** tumours are mostly papillary urothelial neoplasm of low malignant potential (PUNLMP) or low grade. Treat by transurethral resection (TURBT) and fulgurate the base; follow with US. **A recurrence should prompt a search for a syndrome** (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer **2–4%**, aggressive, and **not increased by augmentation**. Screening asymptomatic patients is ineffective, so evaluate haematuria promptly.

Benign lesion	Key point	Treatment
Inflammatory myofibroblastic tumour	ALK rearrangement; intermediate malignancy	Complete resection
Nephrogenic adenoma	Follows trauma, infection, surgery or transplant	TURBT and treat the cause; recurs
Eosinophilic cystitis	Biopsy essential; blood eosinophilia often absent	Steroids, antihistamines
Haemangioma	Severe haematuria; Klippel–Trenaunay, Sturge–Weber	Propranolol, laser; partial cystectomy

6 · Testicular tumours

Why this matters: prepubertal tumours are mostly benign, so the testis can often be spared.

- ~1/100,000 under 15 y, peaking at 2–4 y and puberty; 38–74% benign.
- Before puberty, **teratoma** is the commonest tumour (benign) and **yolk sac tumour (YST)** the commonest malignant one.
- Work-up: Doppler US and lactate dehydrogenase (LDH), **α-fetoprotein (AFP)** and β-human chorionic gonadotrophin (β-hCG).
- US cannot reliably separate benign from malignant. Cystic with echogenic foci suggests teratoma; **"onion-skin" suggests epidermoid cyst**.

	AFP	β-hCG
Source	**YST invariably**	Embryonal carcinoma, mixed tumours; rare before puberty
Half-life	**5 days**	**24 h**
Caveat	Normally high in infancy; reaches adult level (< 10 ng/mL) by **6–9 months**	—

ALGORITHM

1. Use an inguinal approach.
2. If markers are normal and US looks benign, do a **testis-sparing partial orchidectomy** with frozen section.
3. If benign, close.
4. If frozen section shows germ cell malignancy, or AFP is raised, do a radical inguinal orchidectomy.

- Prepubertal teratoma: partial orchidectomy with **no follow-up**. † Postpubertal teratoma is malignant, so do a radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** are stage I. Stage with CT; follow AFP decline by half-life. Stage I: **surveillance, no RPLND or chemotherapy**. Scrotal violation makes it stage II (chemotherapy, no hemiscrotectomy).
- Nodes with normal markers need biopsy; nodes with raised markers mean stage III. Stage III–IV get cisplatin, etoposide and bleomycin. A persistent mass needs biopsy or RPLND.
- **Stromal** tumours: Leydig (commonest; precocious puberty; Reinke crystals ~40%), Sertoli and juvenile granulosa (benign in infants). Do a partial orchidectomy if markers are normal, but radical after Tanner 5. Chemotherapy is rarely needed.

- **Testicular adrenal rests** (poorly controlled congenital adrenal hyperplasia, CAH): increase steroids first; partial orchidectomy if function is threatened.
- Microlithiasis needs no surveillance in childhood.

Disorders of sex development (DSD)

Tumour risk rises with Y material in dysgenetic gonads. High risk: complete gonadal dysgenesis (~35%), partial (~24%), Frasier, Denys–Drash, partial androgen insensitivity (PAIS) and Turner with Y. Intermediate: complete androgen insensitivity (CAIS) and mixed gonadal dysgenesis. Low: no Y. Gonadoblastoma is premalignant, often bilateral, and progresses to dysgerminoma. Gonadectomy is individualised.

> **KEY POINT**
> Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

7 · Other pelvic and genital tumours

Why this matters: these lesions threaten bladder function or mimic RMS.

- **Sacrococcygeal teratoma**: ~1/40,000, mostly girls; neurogenic bladder in 36–51%, worse with Altman II–IV. Renal–bladder US yearly until toilet training; urodynamics for higher types.
- The **Currarino triad** (anterior sacral mass, sacral defect, anorectal malformation) causes neurogenic bladder.
- Non-RMS paratesticular masses (e.g. cystadenoma in von Hippel–Lindau): spare the testis, but inguinal approach and frozen section are mandatory.
- Ovary: neonatal cysts regress. A Sertoli–Leydig tumour should prompt **DICER1 screening**.
- **Urethral prolapse** mimics RMS (a doughnut swelling, in prepubertal Black girls); treat with topical oestrogen or steroid.

★ Putting it together: common paediatric decisions

A · VUR after febrile UTI

Most reflux needs no surgery.

ALGORITHM

1. Renal–bladder ultrasound (US); voiding cystourethrogram (VCUG) if febrile urinary tract infections (UTIs) recur or need admission, or US is abnormal.
2. Treat bladder and bowel dysfunction (BBD), constipation first.
3. Low risk (grade I–II, no BBD, boy, older): observe with or without continuous antibiotic prophylaxis (CAP). High risk (grade ≥ III, girl, BBD, scars, infant): CAP.
4. Breakthrough UTI, new scars, or persistent high grade with poor function: correct the vesicoureteral reflux (VUR) by injection or reimplant (*see Ch 41, 53*).

B · Dilated upper tract

Operate only for these triggers.

Condition	Key test	Operate when	Operation
Pelviureteric junction (PUJ) obstruction	Mercaptoacetyltriglycine (MAG3) diuretic renogram	Function falling or < 40%, worsening dilatation, symptoms, infection	Anderson–Hynes pyeloplasty
Primary megaureter	US, MAG3, VCUG	Falling function, infection, progression; most resolve	Tapered/folded reimplant; infants: diversion or balloon
Ureterocele / ectopic ureter	US, VCUG, functional scan	Obstruction, infection, incontinence (girls)	Incision, upper-pole heminephrectomy or ureteroureterostomy (*see Ch 52*)
Multicystic dysplastic kidney	US and DMSA (dimercaptosuccinic acid) scan (non-functioning)	Rarely; most involute	Observe; check the other kidney for VUR/PUJ

C · Undescended testis

Early orchidopexy protects germ cells.

ALGORITHM

1. **Palpable**: orchidopexy at **6–18 months** (ideally by 12); no imaging.
2. **Non-palpable**: examination under anaesthesia, then laparoscopy. Blind-ending vessels: stop. Vessels and vas entering the ring: groin exploration. Intra-abdominal testis: laparoscopic orchidopexy (primary/Fowler–Stephens).
3. **Bilateral non-palpable, or undescended testis with hypospadias**: disorder of sex development (DSD) work-up first *(see Ch 59)*.

D · Acute scrotum

Never let a scan delay exploring torsion.

	Testicular torsion	Appendage torsion	Epididymitis
Typical	Sudden pain, vomiting, high-riding testis	Prepubertal; upper-pole tenderness, "blue dot"	Gradual; urinary symptoms; seek an anomaly
Action	**Explore at once** (Doppler misses partial torsion); fix both sides, remove if non-viable	Analgesia, rest	Culture; antibiotics if infected

E · Newborn with atypical genitalia

Missed salt-wasting can kill in week one.

> **KEY POINT**
> Bilateral impalpable gonads with atypical genitalia = salt-wasting congenital adrenal hyperplasia (CAH) until proven otherwise. Check electrolytes, 17-hydroxyprogesterone (after 36 h) and karyotype; crisis at days 5–7. Sex assignment and surgery are not emergencies (multidisciplinary team).

F · Wilms tumour vs neuroblastoma

Age, imaging and markers separate them.

	Wilms tumour	Neuroblastoma
Age	Median 3.5 years	† Younger (median ~18 months)
Imaging	Intrarenal († claw sign); may enter the inferior vena cava (IVC)	Calcification, vessel encasement; † crosses the midline, displaces the kidney
Markers / genetics	WT1, 11p15; WAGR, Denys–Drash, Beckwith–Wiedemann	Urinary vanillylmandelic and homovanillic acid (VMA/HVA) raised in 90–95%; MYCN amplification = high risk
Strategy	COG (North America): nephrectomy first · SIOP (Europe): chemotherapy first	By INRG risk: observation, then surgery, then multimodal therapy